GeoGebra Interactive Physics Fundamentals Framework (GIPFF)

Companion Guide

Contents

GeoGebra Interactive Physics Fundamentals Framework (GIPFF)

Volume 1: Mechanics

This GeoGebra Classroom contains interactive physics simulation applets developed to support conceptual understanding, visualization, and inquiry-based learning in Mechanics.

This framework is intended to provide an accessible collection of interactive learning activities for students, educators, and anyone interested in learning fundamental physics.

The applets are primarily based on topics discussed in the following reference books:

- Physics for Scientists and Engineers — Raymond A. Serway, John W. Jewett
- Physics for Scientists and Engineers — Douglas C. Giancoli
- University Physics — Hugh D. Young, Roger A. Freedman, A. Lewis Ford
- Fundamentals of Physics — David Halliday, Robert Resnick, Jearl Walker

This framework emphasizes exploratory learning through interactive visualization. Rather than relying heavily on step-by-step procedural instructions, the activities are designed to encourage users to investigate physical concepts through observation, interaction, experimentation, and independent thinking.

Users are encouraged to freely manipulate variables, explore different physical situations, compare graphical and mathematical representations, and develop deeper conceptual intuition in physics.

GIPFF is provided free of charge for educational use.

Learning Approach

Each activity follows the following structure:

- **Learning Focus** – identifies the main concepts to investigate.
- **Interactive Applet** – explore the simulation by changing variables and observing the results.
- **Key Observations** – highlights important physical relationships revealed by the applet.
- **Challenge / Question** – encourages deeper thinking and conceptual understanding.

There is no single prescribed way to explore an applet. Users are encouraged to investigate freely, test predictions, compare different situations, and construct their own understanding through exploration.

Applet Interface & Controls

- **Sliders:** The most common interactive objects in these applets are sliders. You can adjust a slider's value using your mouse pointer, or use the keyboard arrow keys for more precise control.

- **Fixed Points:** A tiny solid circle (•) represents a fixed point that cannot be moved directly.

- **Movable Points:** A tiny hollow circle (◦) indicates a movable point. It can represent the initial or terminal point of a vector, a coordinate on a graph, part of an object, or a contact point with another object. Moving the point changes the position or properties of the object associated with it.

- **Animation Controls:** Use the **Animate** button to start the simulation, **Stop** to pause it, and **Reset** to return to the initial conditions. These three buttons are often paired with a driving slider (such as time, angle, or angular displacement), allowing you to manually control the progression of the animation.

- **Calculated Values:** To help you verify calculations, most physical quantities—such as angle, vector magnitude, vector components, position, velocity, acceleration, force, distance, maximum height, speed, energy, and momentum—are displayed to 5 significant figures. For applets requiring gravitational calculations, the acceleration due to gravity is implicitly set to $g = 9.80\ m/s^2$.

Unit Conversion

Activity: Unit Conversion

Learning Focus

- Understand the principle of unit conversion using conversion factors.
- Observe how units cancel systematically in calculations.
- Follow step-by-step transformation between different unit systems.

Interactive Applet

Key Concept in Unit Conversion:
Write the conversion factor as a ratio of 1:

e.g., 1 km = 1000 m is written as $1 = \frac{1000\,\mathrm{m}}{1\,\mathrm{km}}$ or $1 = \frac{1\,\mathrm{km}}{1000\,\mathrm{m}}$

Choose the ratio so the units cancel appropriately.

Example 1: Force Conversion
The thrust of a rocket engine is 686 N. Convert this to pounds-force (lbf).
(1 lbf = 4.448 N)

Solution:

$$686\,\cancel{\mathrm{N}} \times \frac{1\,\mathrm{lbf}}{4.448\,\cancel{\mathrm{N}}} = 154\,\mathrm{lbf}$$

Example 2: Area Conversion
Express an area of $2.40\,\mathrm{m}^2$ in square centimeters (cm^2). ($1\,\mathrm{m} = 100\,\mathrm{cm}$)

Solution:

$$2.40\,\mathrm{m}^2 \times \left(\frac{100\,\mathrm{cm}}{1\,\mathrm{m}}\right)^2 = 2.40\,\cancel{\mathrm{m}^2} \times \frac{10,000\,\mathrm{cm}^2}{1\,\cancel{\mathrm{m}^2}} = 2.40 \times 10^4\,\mathrm{cm}^2$$

Step

Key Observations

- A conversion factor is always equal to 1 in value.
- Proper unit cancellation ensures correct physical interpretation.
- The same quantity can be represented in multiple unit systems.

Challenge / Question

Why must every valid conversion factor be equivalent to 1, even though its numerical form changes?

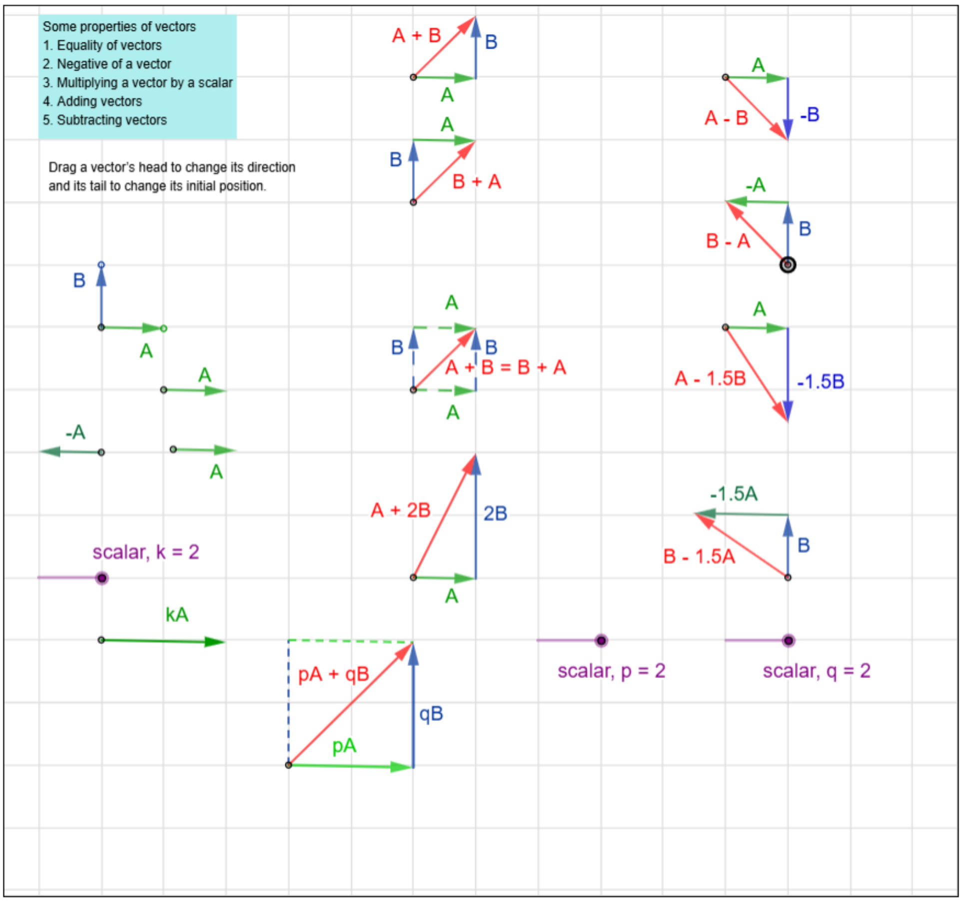
Vectors

Activity: Some Properties of Vectors

Learning Focus

- Explore fundamental properties of vectors through graphical representation and interaction.
- Observe how scalar multiplication affects vector magnitude and direction.
- Investigate how linear combinations of vectors generate new vectors.

Interactive Applet



Key Observations

- Equal vectors have the same magnitude and direction even when located at different positions.
- Scalar multiplication changes the magnitude of a vector and either maintains or reverses its direction depending on whether the scalar is positive or negative.
- Vector addition and subtraction can be interpreted graphically using head-to-tail constructions.

Challenge / Question

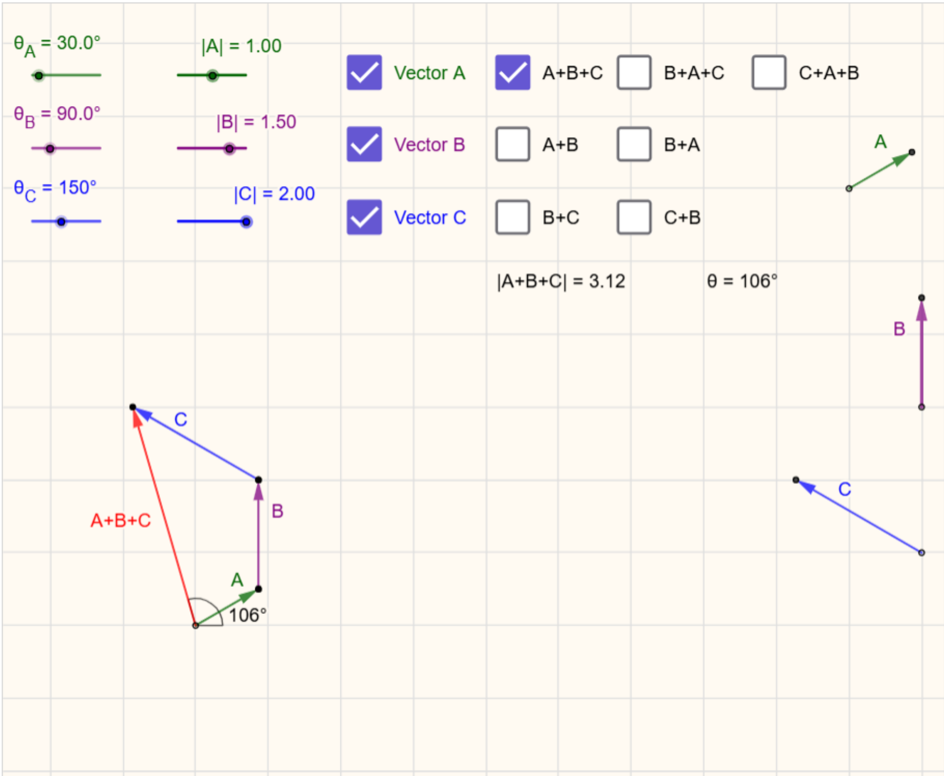
Why can two vectors located at different positions still represent the same vector quantity?

Activity: Adding 3 Vectors Graphically

Learning Focus

- Explore graphical vector addition using the head-to-tail method.
- Observe how vector magnitude and direction affect the resultant vector.
- Compare different vector addition sequences graphically.

Interactive Applet



Key Observations

- Graphical vector addition can be performed using the head-to-tail method.
- Vector addition obeys the commutative property (**A** + **B** = **B** + **A**) as well as the associative property ((**A** + **B**) + **C** = **A** + (**B** + **C**)).
- Because of these properties, the final resultant vector is entirely unaffected by the order or grouping of the individual vectors.

Challenge / Question

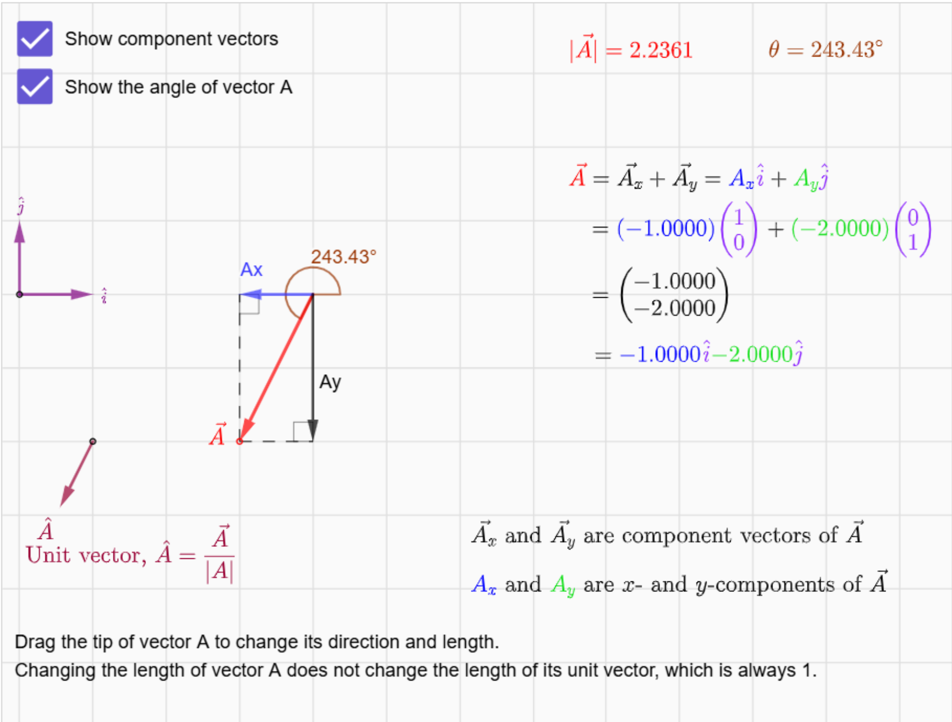
Why does changing the order of vector addition not affect the final resultant vector?

Activity: Unit Vectors

Learning Focus

- Explore the relationship between a vector and its unit vector.
- Take a look at how a vector is broken down into its component vectors.
- Notice how these component vectors are written in unit-vector notation, and how adding them together gives you the original vector.

Interactive Applet



Key Observations

- A unit vector has magnitude 1 and indicates direction only.
- Changing the magnitude of vector **A** changes its components but does not change the length of its unit vector.
- A vector can be expressed as the sum of scalar multiples of unit vectors **i** and **j** along coordinate axes.

Challenge / Question

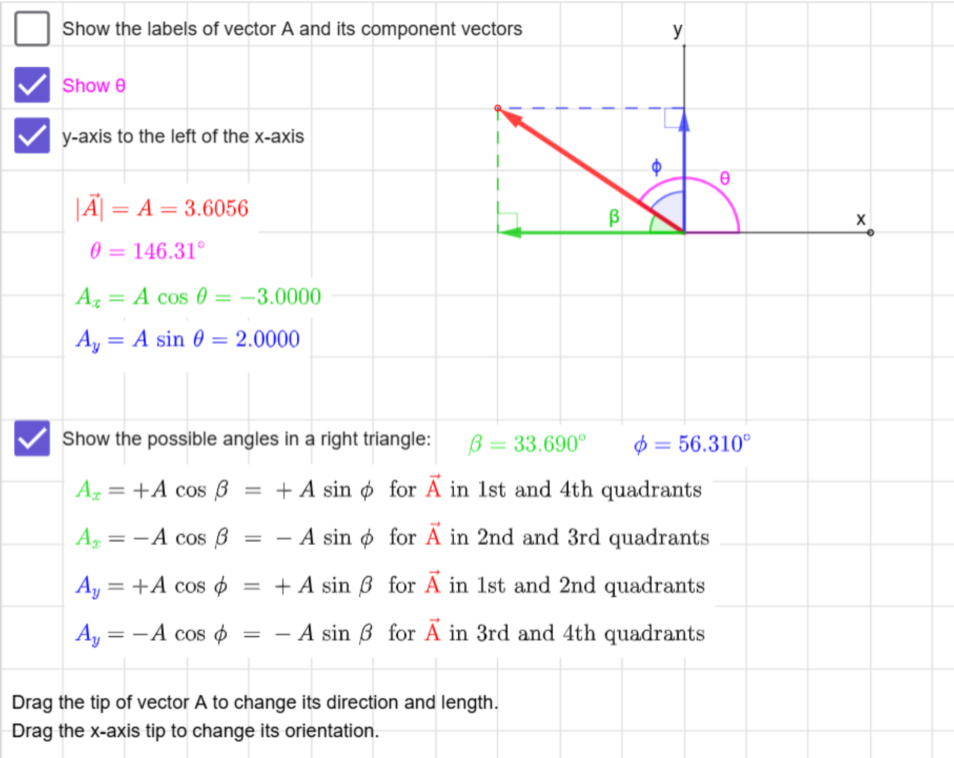
Why does changing the magnitude of vector **A** not affect the length of its unit vector?

Activity: Vector Components

Learning Focus

- Explore how a vector can be resolved into x- and y-components.
- Observe how vector components depend on the choice of coordinate axes.
- Relate vector components to right-triangle geometry and angle measurements.

Interactive Applet



Key Observations

- The values of vector components depend on the orientation of the coordinate axes.
- A vector has different component values in different coordinate systems.
- These components can be determined geometrically using the interior angles of the right triangle.

Challenge / Question

Why can the same vector have different component values when the coordinate axes are rotated, even though the vector itself remains unchanged?

Activity: Vector Addition: Worked Example

Learning Focus

- Observe how vectors are broken down into horizontal (x) and vertical (y) components using trigonometric functions.
- Track how the individual x-components and y-components are added together independently to find the resultant components.
- Notice how the total components are recombined using the Pythagorean theorem for magnitude and the tangent function for direction.

Interactive Applet

Example: Displacement of a Moving Object on a 2D Plane

A delivery robot moves on a flat warehouse floor starting from point A. Its path is:
100 m at 30° north of east, then 150 m due north, and finally 200 m at 60° west of north.
Find the resultant displacement of the robot from point A.

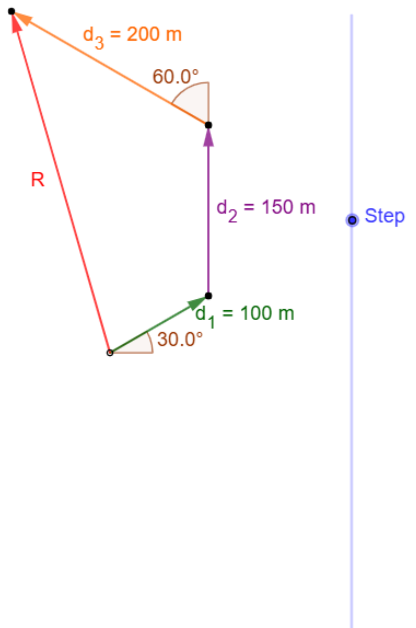
Solution:

Step 1: Sketch the displacement vectors

Let east be $+x$ and north be $+y$.

Step 2: Resolve each displacement into components

$$\vec{d_1} = \vec{d}_{1x} + \vec{d}_{1y} = \begin{pmatrix} 100 \text{ m } \cos 30^\circ \\ 100 \text{ m } \sin 30^\circ \end{pmatrix} = \begin{pmatrix} 86.6 \\ 50 \end{pmatrix} \text{m}$$



Key Observations

- Vectors cannot be added by simply summing their magnitudes unless they point in the exact same direction; they must be combined component-by-component.
- The mathematical signs (+ or -) of the components depend strictly on which quadrant each vector points toward.
- The example explicitly steps through how to combine the calculated components to find the final overall magnitude and angle of the resultant vector.

Challenge / Question

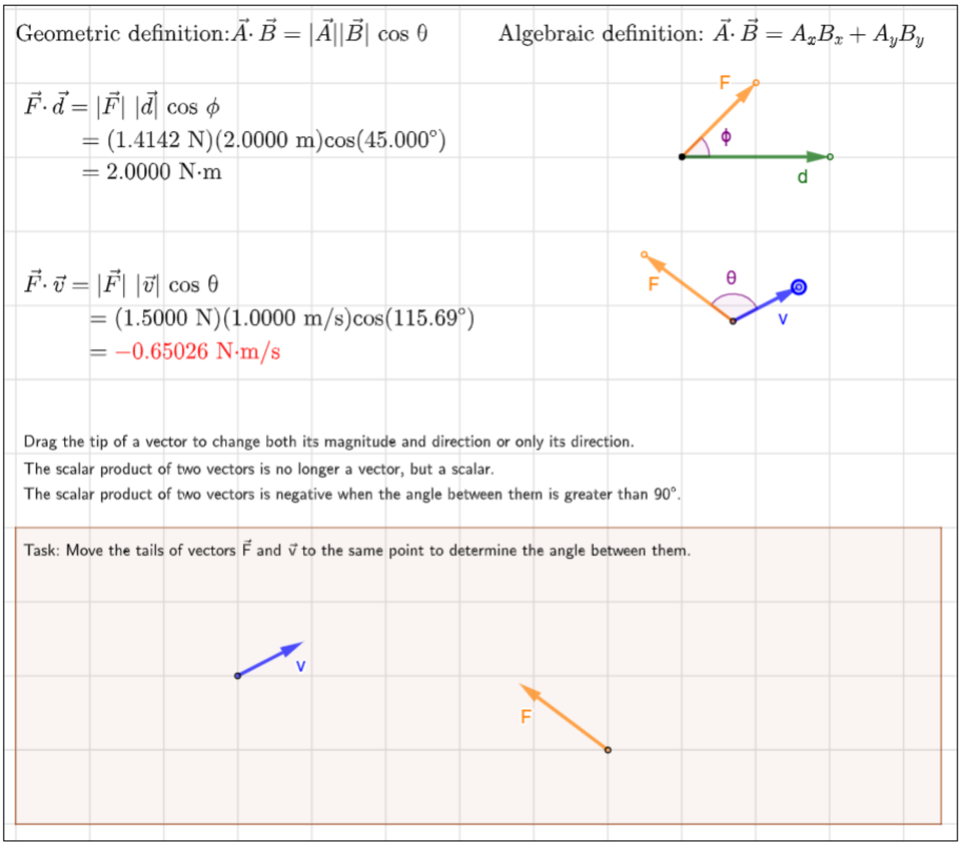
Why is it incorrect to just add the magnitudes of the three initial vectors together to get the resultant magnitude?

Activity: Dot Product (Scalar Product)

Learning Focus

- Explore the geometric meaning of the dot product between two vectors.
- Compare how the dot product behaves for acute and obtuse angles.
- Resolve the vectors into components to verify that the algebraic definition yields the same result as the geometric definition.

Interactive Applet



Key Observations

- The dot product depends on both the magnitudes of vectors and the cosine of the angle between them.
- The dot product is positive for angles less than 90°, zero at 90°, and negative for angles greater than 90°.
- Correctly positioning vectors tail-to-tail allows the angle between them to be directly visualized and measured.

Challenge / Question

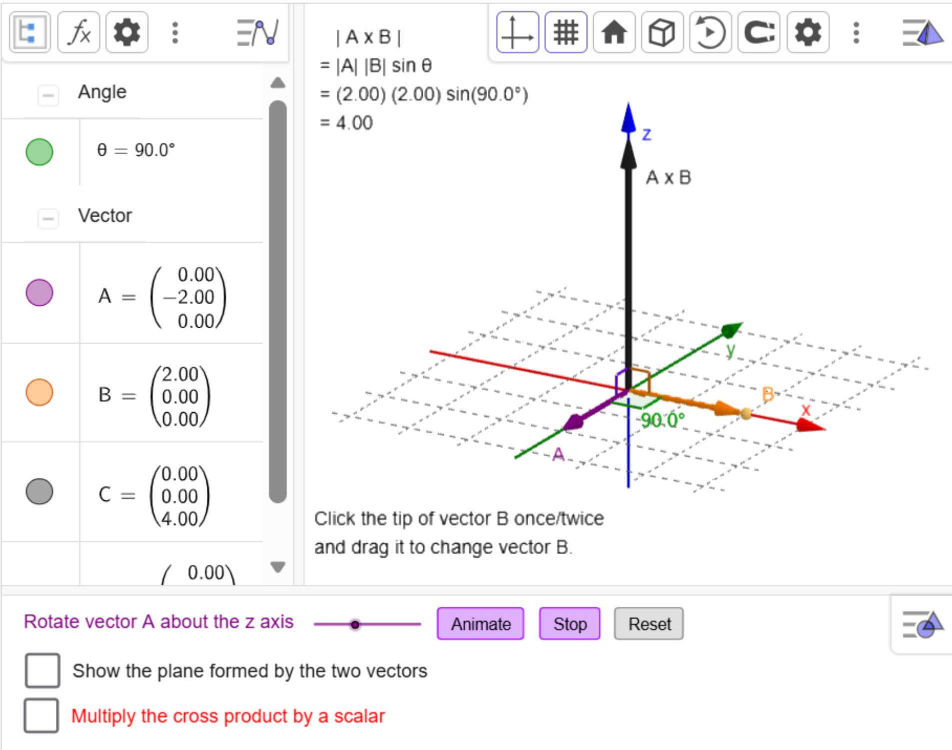
Why does the dot product become negative when the angle between two vectors exceeds 90°?

Activity: Cross Product (Vector Product)

Learning Focus

- Explore the geometric meaning of the cross product between two vectors in three dimensions.
- Observe how the magnitude and direction of **A** x **B** depend on both the magnitudes and relative orientations of the vectors.
- Investigate the effect of scalar multiplication on the resulting cross-product vector.

Interactive Applet



Key Observations

- The cross-product vector is perpendicular to both **A** and **B**.
- The magnitude of the cross product depends on the magnitudes of the vectors and the sine of the angle between them.
- Multiplying the cross-product vector by a scalar changes its magnitude and may reverse its direction.

Challenge / Question

Why does the cross product become zero when the two vectors are parallel, even if their magnitudes are large?

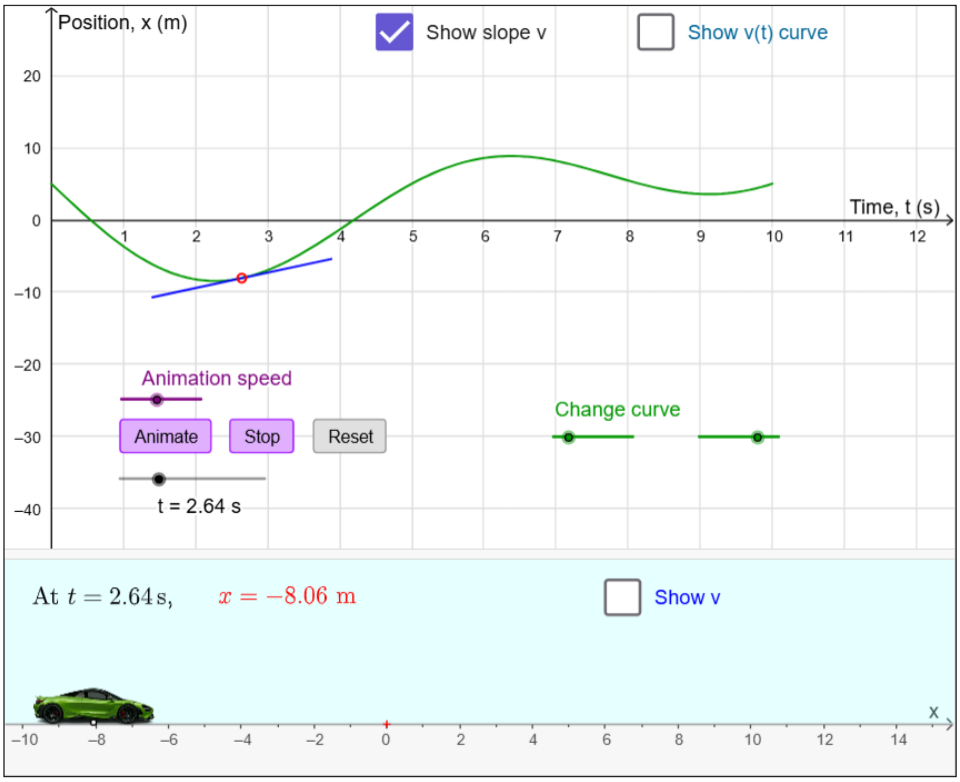
One-Dimensional Motion

Activity: Position and Velocity in 1D Motion

Learning Focus

- Relate the position–time graph to the actual movement of the car along the x-axis.
- Explore the relationship between position and velocity in one-dimensional motion.
- Observe how the slope of a position–time graph affects the motion of an object.

Interactive Applet



Key Observations

- The slope of the position–time graph represents the velocity of the object.
- A steeper graph corresponds to a larger magnitude of velocity.
- Positive and negative slopes indicate motion in opposite directions along the x-axis.

Challenge / Question

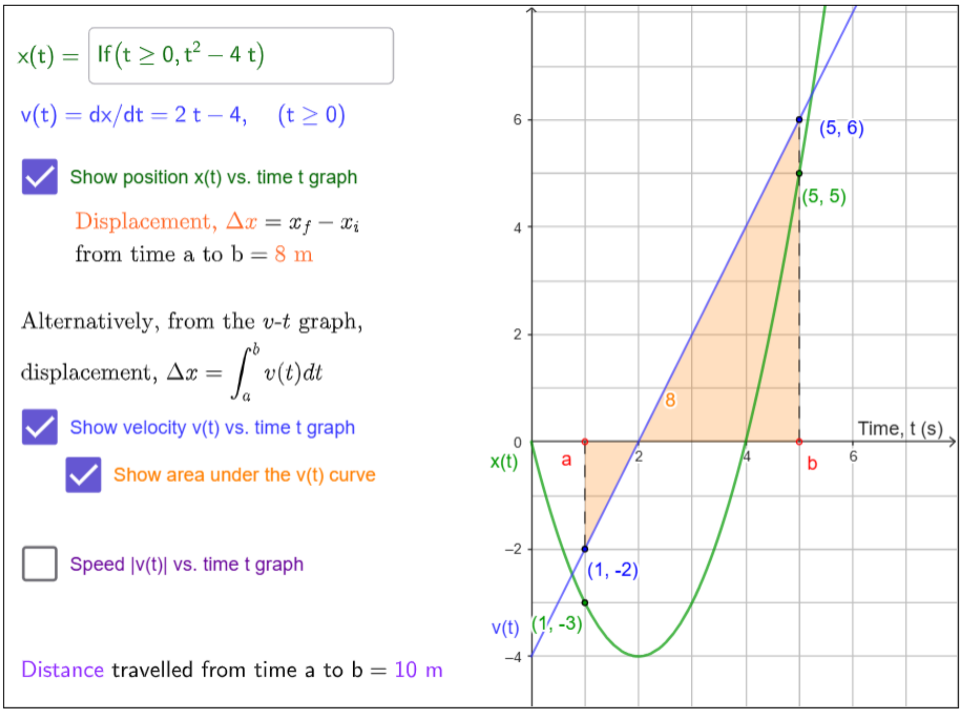
Where on a position-time graph does the car momentarily stop?

Activity: Finding Displacement from x-t and v-t Graphs

Learning Focus

- Explore the relationship between position–time and velocity–time graphs.
- Observe how displacement between two times can be determined graphically.
- Investigate how changing the position function affects the corresponding velocity graph and displacement.

Interactive Applet



Key Observations

- Velocity is the slope of the position–time graph.
- A positive area under a velocity–time graph represents forward displacement, while a negative area represents backward displacement.
- The displacement between two times is the sum of the signed areas under the velocity–time graph.

Challenge / Question

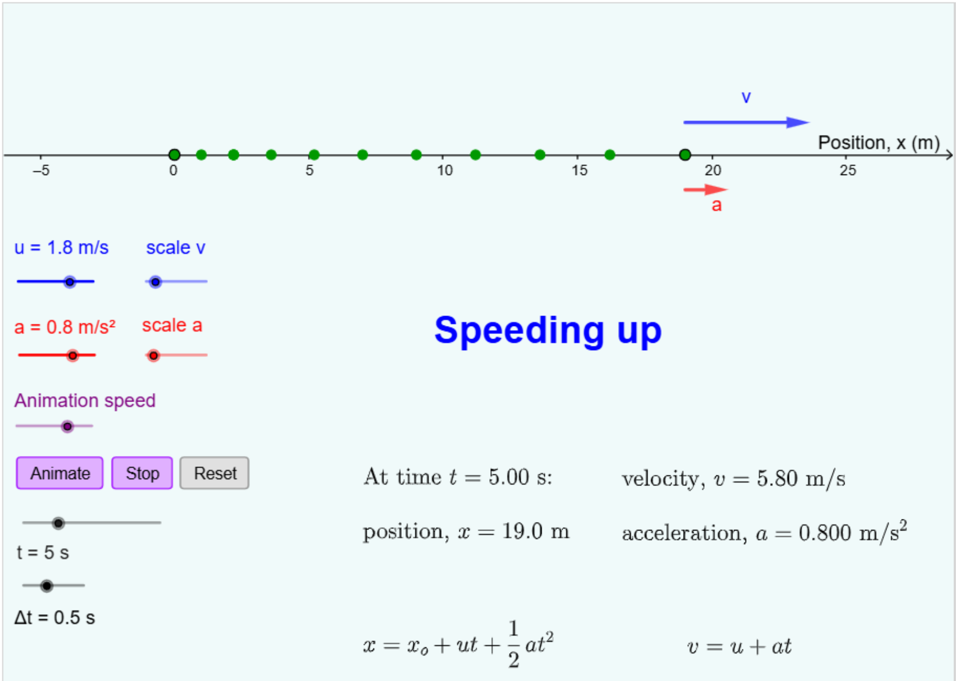
Why can the total distance traveled be larger than the displacement obtained from the area under the velocity–time graph?

Activity: 1D Motion with Constant Acceleration (Motion Diagram)

Learning Focus

- Explore one-dimensional motion with constant acceleration using motion diagrams.
- Observe how velocity changes with time under positive, negative, and zero acceleration.
- Relate spacing between successive positions to the speed of the moving object.

Interactive Applet



Key Observations

- Increasing spacing between adjacent dots indicates increasing speed. The velocity and acceleration vectors are in the same direction.
- Decreasing spacing between adjacent dots indicates decreasing speed. The velocity and acceleration vectors are in opposite directions.
- A constant acceleration vector implies that velocity changes linearly with time.

Challenge / Question

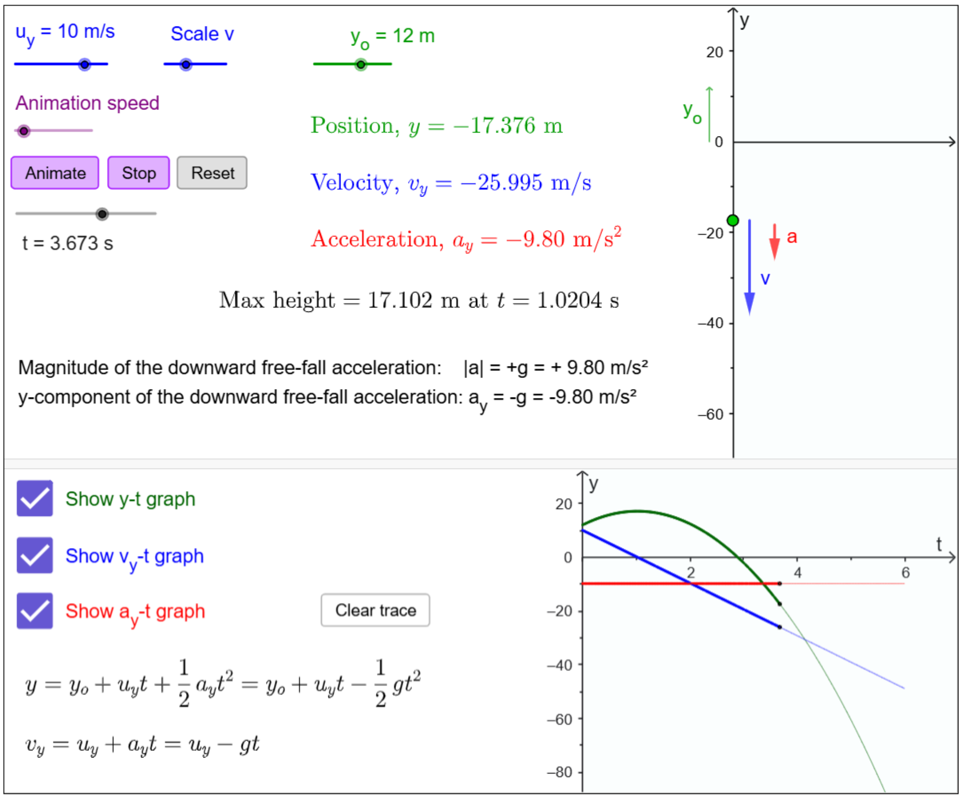
How can an object have negative acceleration while still moving in the positive x-direction?

Activity: 1D Free Falling

Learning Focus

- Explore one-dimensional motion under constant gravitational acceleration.
- Observe how position and velocity change during free fall.
- Investigate the effects of initial height and initial velocity on the motion of the object.

Interactive Applet



Key Observations

- The acceleration due to gravity remains constant throughout the motion.
- The speed decreases during upward motion and increases during downward motion.
- At maximum height, the velocity becomes zero momentarily while acceleration remains nonzero.

Challenge / Question

Why does the object still have acceleration at the maximum height even though its velocity is zero at that instant?

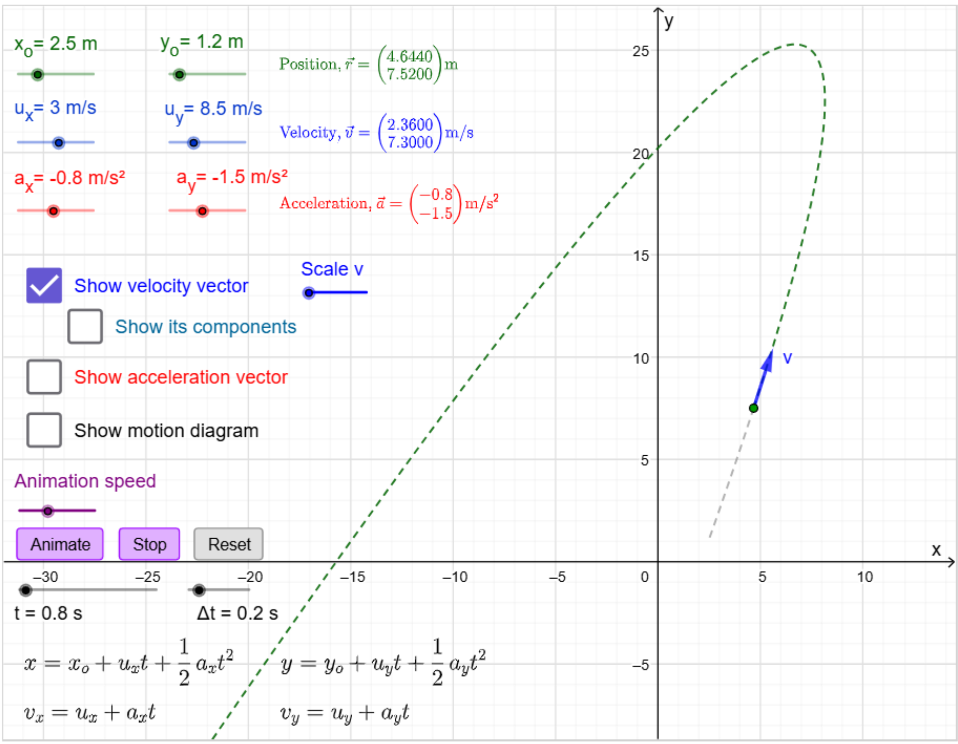
Two-Dimensional Motion

Activity: 2D Motion with Constant Acceleration

Learning Focus

- Explore two-dimensional motion under constant acceleration.
- Observe how the x- and y-components of position and velocity change independently with time.
- Investigate how initial position, initial velocity, and acceleration affect the trajectory of an object.

Interactive Applet



Key Observations

- Constant acceleration continuously changes the velocity vector with time.
- Different combinations of initial velocity and acceleration produce different trajectories in the x–y plane.
- Two-dimensional motion can be analyzed as two independent one-dimensional motions linked only by time.

Challenge / Question

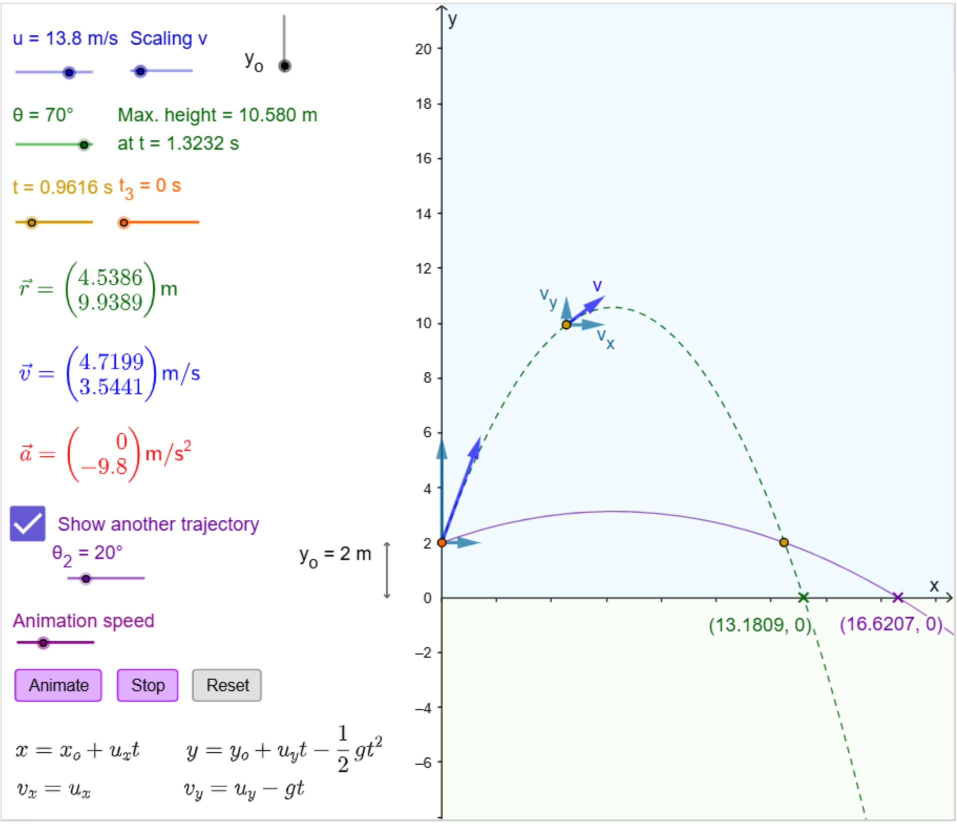
How can an object move along a curved trajectory even though the acceleration vector remains constant?

Activity: Projectile Motion

Learning Focus

- Explore two-dimensional projectile motion under constant gravitational acceleration.
- Observe how the position and velocity change during the motion.
- Investigate how launch speed, launch angle, and initial height affect the projectile trajectory and horizontal range.

Interactive Applet



Key Observations

- The horizontal and vertical components of projectile motion are independent of each other.
- The acceleration vector remains constant and vertically downward throughout the motion.
- For the same launch speed and zero initial height, two projection angles whose sum is 90° produce the same horizontal range.

Challenge / Question

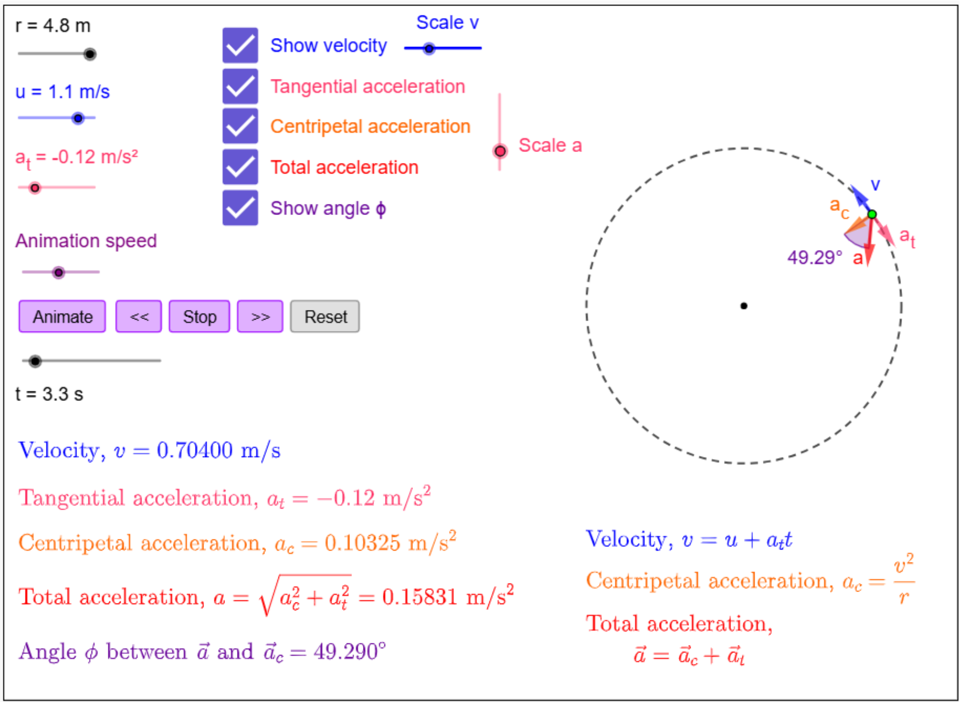
Why can two different launch angles produce the same horizontal range even though the trajectories are different?

Activity: Circular Motion

Learning Focus

- Explore circular motion with tangential acceleration.
- Observe how tangential acceleration affects the speed, centripetal acceleration, and total acceleration of an object.
- Investigate how radius, speed, and tangential acceleration influence motion along a circular path.

Interactive Applet



Key Observations

- The velocity vector is always tangent to the circular path.
- The centripetal acceleration vector always points toward the center of the circular path.
- The total acceleration is the vector combination of tangential acceleration and centripetal acceleration.

Challenge / Question

How can an object accelerate even when moving at constant speed in a circular path?

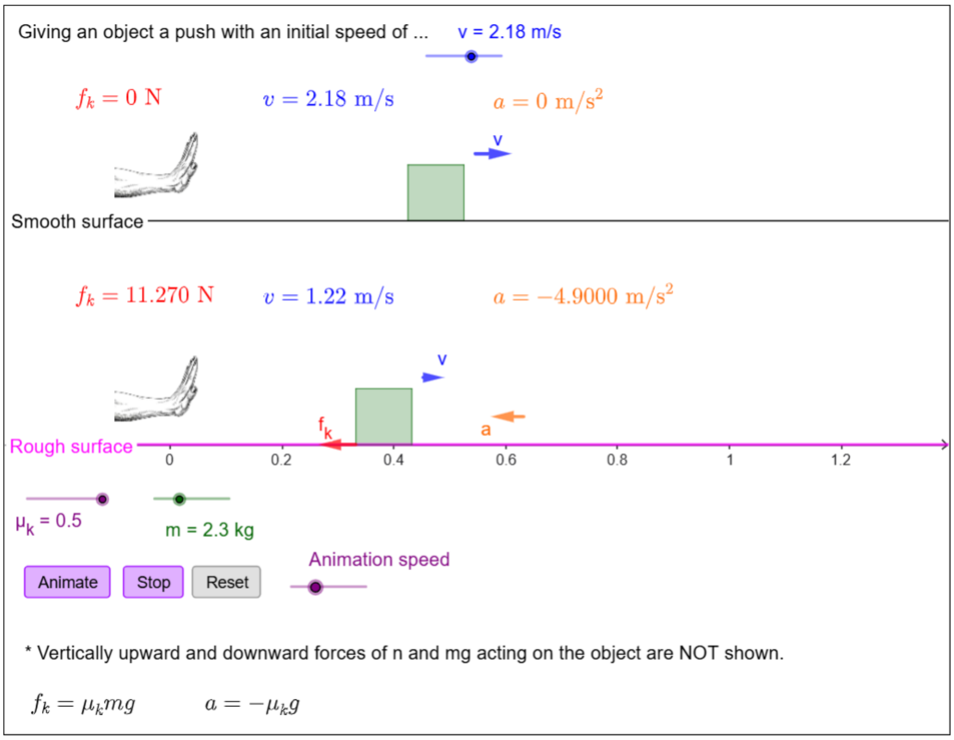
Newton's Laws of Motion

Activity: Giving an Object an Initial Push

Learning Focus

- Explore the motion of objects on smooth and rough horizontal surfaces after receiving an initial push.
- Observe how kinetic friction affects the velocity and acceleration of a moving object.
- Investigate how mass and the coefficient of kinetic friction influence the frictional force and the acceleration of the object.

Interactive Applet



Key Observations

- On a smooth surface, an object continues to move at a constant velocity because there are no horizontal forces to change its state of motion.
- Kinetic friction causes an acceleration opposite to the direction of motion on the rough surface.
- Increasing the mass increases the frictional force but does not change the acceleration because the extra mass also increases the object's inertia.

Challenge / Question

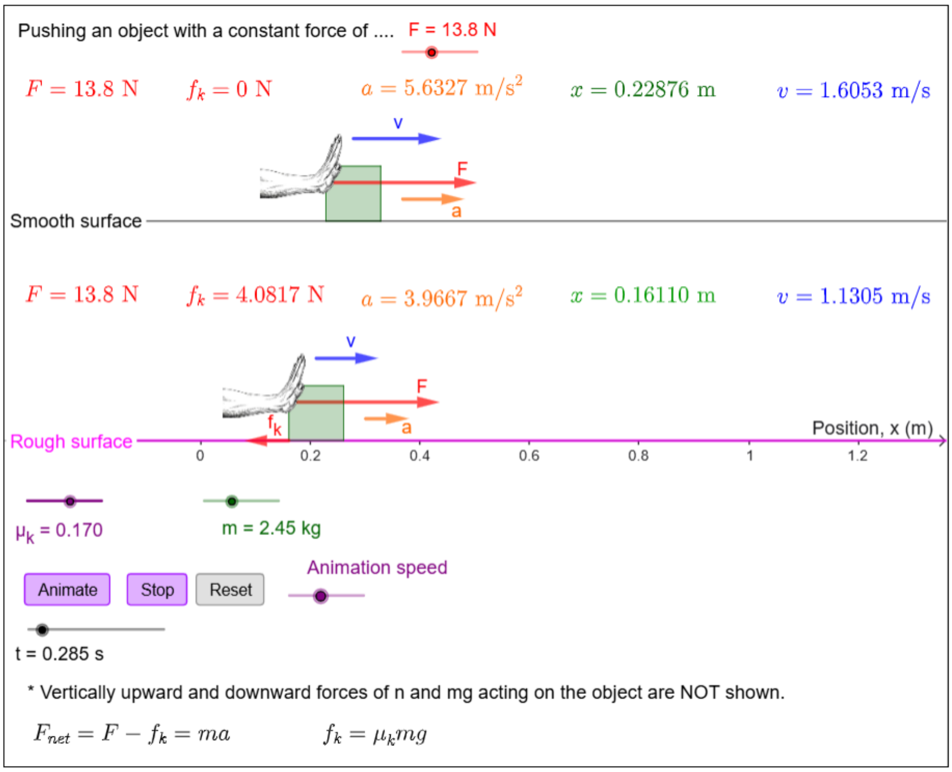
Why does the object on the smooth surface continue moving even though no horizontal force acts on it after the push?

Activity: Pushing an Object

Learning Focus

- Explore the motion of objects subjected to a constant horizontal pushing force.
- Observe how kinetic friction affects the net force and acceleration of an object.
- Investigate the relationship between applied force, mass, and surface roughness (μ_k) in determining an object's acceleration.

Interactive Applet



Key Observations

- On the smooth surface, the applied force causes constant acceleration in the direction of the force.
- On the rough surface, kinetic friction acts opposite to the motion and reduces the net force acting on the object.
- Increasing mass decreases acceleration. On smooth surfaces, this is due to inertia ($a = \frac{F}{m}$); on rough surfaces, the effect is magnified because more mass also increases friction.

Challenge / Question

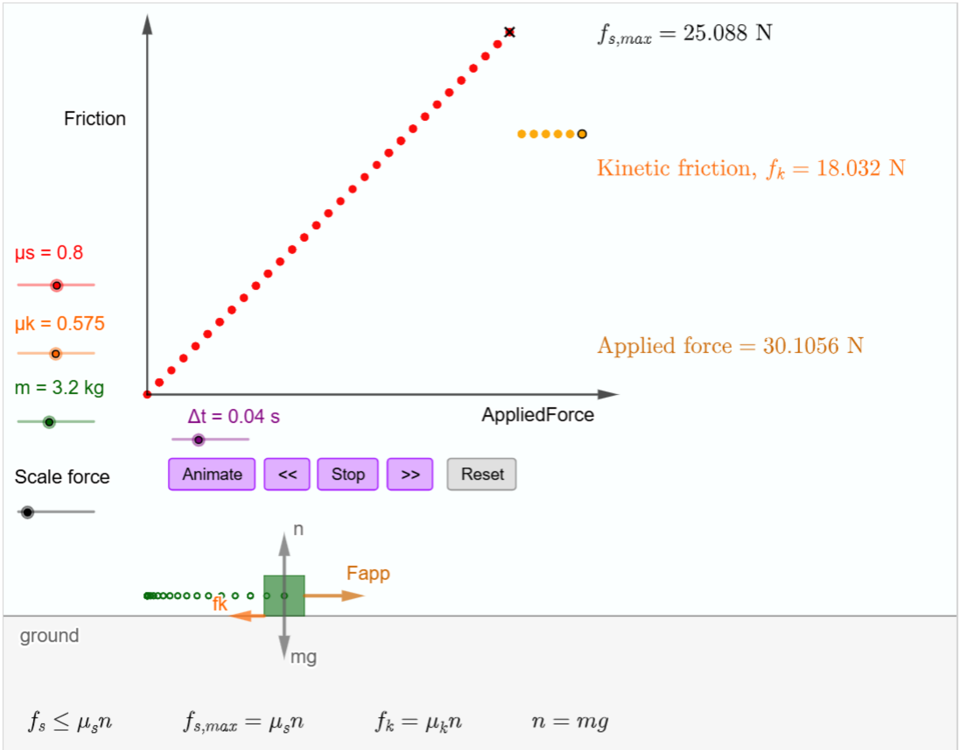
Why can two objects pushed by the same force have different accelerations?

Activity: Static Friction and Kinetic Friction

Learning Focus

- Explore the behavior of static friction and kinetic friction acting on an object.
- Observe how friction changes as the applied force increases from zero to a value exceeding the maximum static friction.
- Investigate how mass and the coefficients of friction determine the thresholds for static and kinetic friction.

Interactive Applet



Key Observations

- Static friction adjusts its magnitude to match the applied force until the maximum static friction is reached.
- Once the applied force exceeds the maximum static friction, the object begins to move and the friction changes to kinetic friction.
- Kinetic friction remains constant regardless of how much further the applied force increases.

Challenge / Question

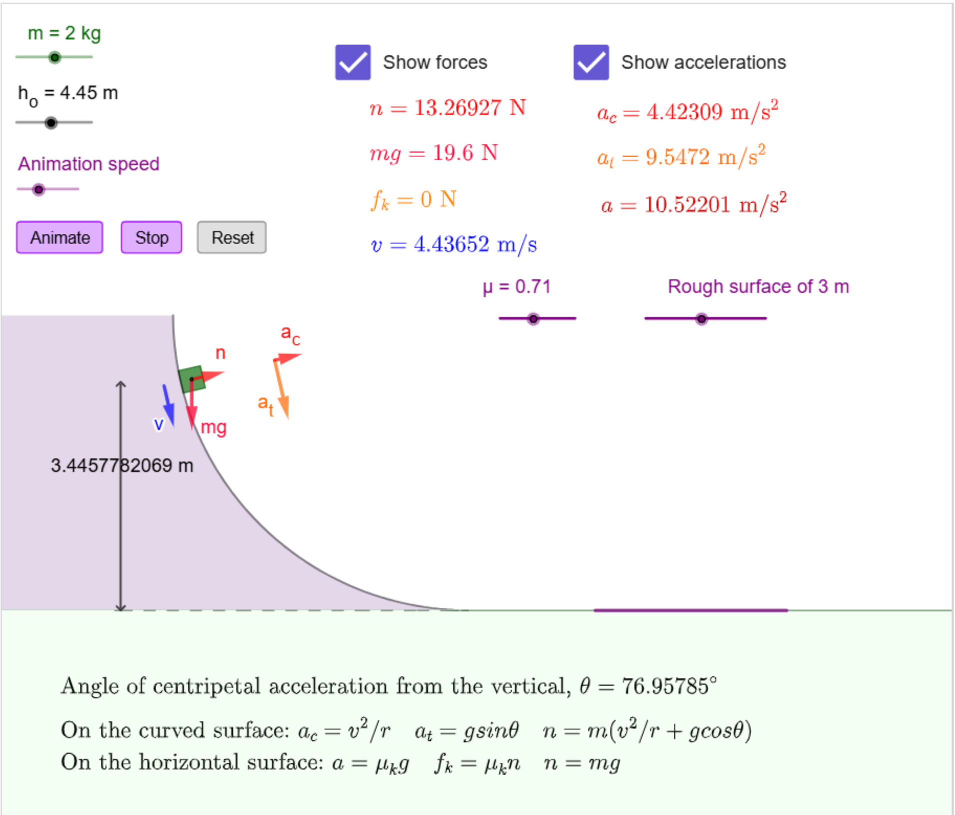
Why does the frictional force suddenly decrease when the box starts moving?

Activity: Forces and Accelerations

Learning Focus

- Explore how forces and accelerations change as an object moves through different regions of motion.
- Observe how gravitational and normal forces affect the accelerations (tangential and centripetal) required to guide the box along the curved surface.
- Investigate how the box's speed upon entering the rough region and the kinetic friction determine its deceleration and whether it stops inside the region.

Interactive Applet



Key Observations

- On the curved frictionless track, the motion involves both tangential acceleration and centripetal acceleration.
- The normal force changes continuously along the curve because it depends on both the changing track angle and the box's speed.
- On the horizontal rough surface, kinetic friction causes a constant acceleration opposite to the motion, causing the box to slow down.

Challenge / Question

What forces (or components of forces) cause the box to accelerate tangentially and centripetally along the curved surface?

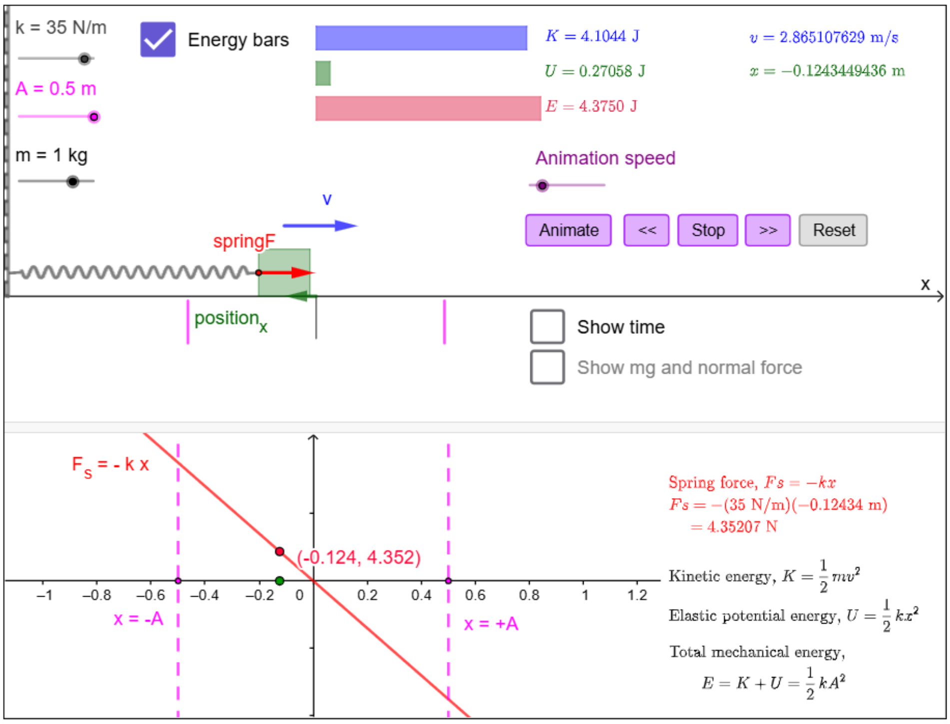
Work and Energy

Activity: Mass-Spring System

Learning Focus

- Explore oscillatory motion in a horizontal mass–spring system on a frictionless surface.
- Observe how displacement, velocity, spring force, and energy change continuously during the oscillation.
- Investigate how the spring constant and amplitude affect the maximum speed and energy of the system.

Interactive Applet



Key Observations

- The spring force always acts toward the equilibrium position and is proportional to the displacement of the box.
- The velocity of the box is maximum at the equilibrium position and zero at the turning points of the motion.
- Kinetic energy and elastic potential energy continuously transform into each other while the total mechanical energy remains constant.

Challenge / Question

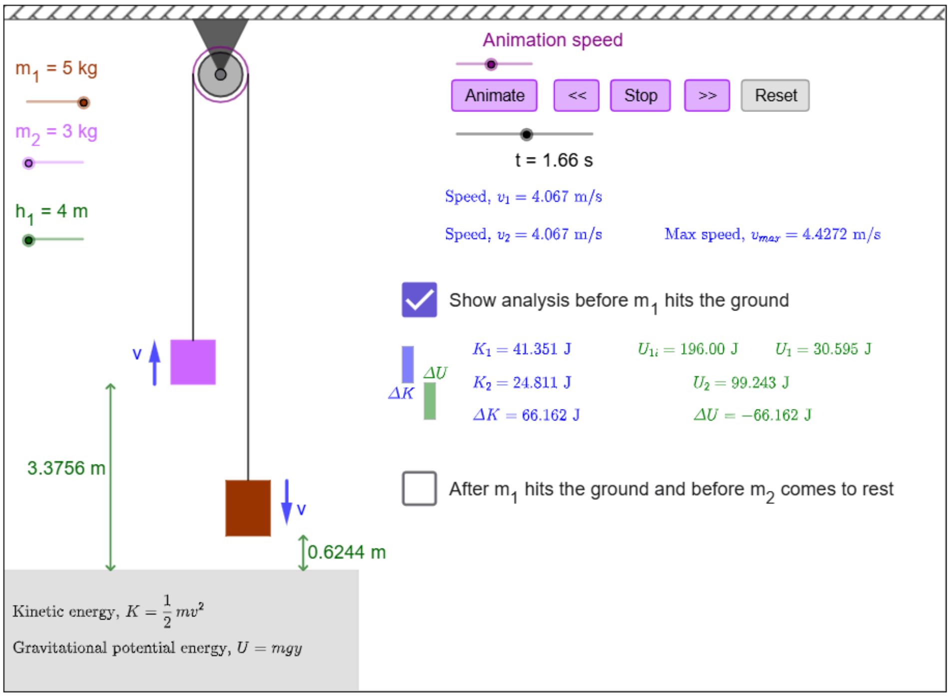
Why is the velocity maximum when the spring force and elastic potential energy are zero?

Activity: Objects over a Pulley

Learning Focus

- Explore the motion of two connected objects moving under the influence of gravity (an Atwood machine).
- Observe the continuous conversion between gravitational potential energy and kinetic energy during the motion.
- Analyze the two distinct phases of motion: before and after the heavier mass strikes the ground.

Interactive Applet



Key Observations

- The total kinetic energy of the two-object system increases as gravitational potential energy decreases.
- The speed of both objects is always the same because they are connected by a light inextensible string.
- After the heavier object reaches the ground, the remaining kinetic energy of the lighter object is converted into additional gravitational potential energy as it continues to rise.

Challenge / Question

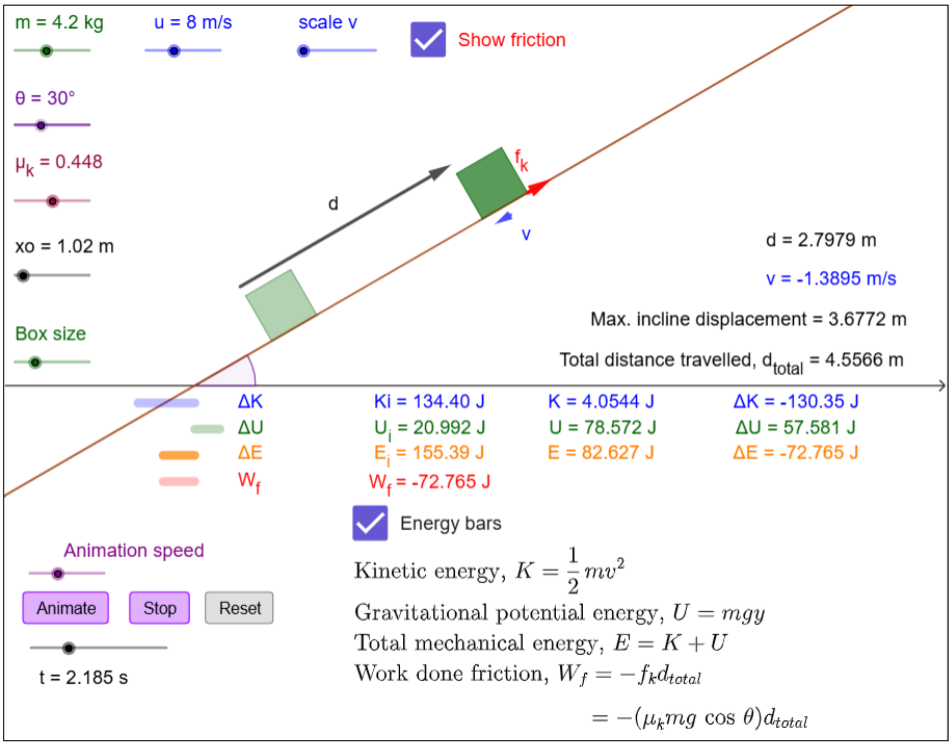
Why does the lighter object continue moving upward after the heavier object has already reached the ground?

Activity: Work done by Friction and Mechanical Energy on an Incline

Learning Focus

- Observe how a rough surface drains total mechanical energy from the system.
- Explore how the cumulative distance traveled by an object—even when it changes direction—influences the work done by friction.
- Understand why work done by friction is always negative and how it leads to an overall decrease in mechanical energy.

Interactive Applet



Key Observations

- Total mechanical energy decreases continuously; this loss is exactly equal to the negative work done by kinetic friction.
- Gravity is path-independent and only depends on vertical height. Friction is path-dependent, continuously draining energy along the entire total distance traveled.
- At peak height, velocity reaches zero. Depending on the surface roughness, the box either comes to a complete stop or the friction force reverses its direction to oppose a downward slide.

Challenge / Question

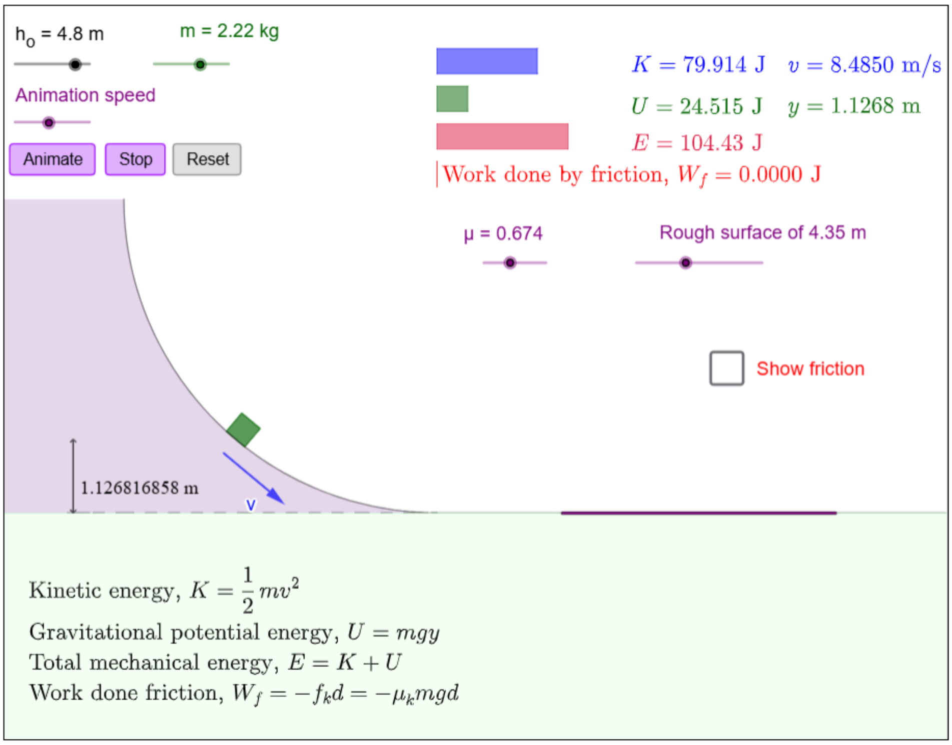
When the block slides up and then back down to its starting point, will its final kinetic energy be less than, equal to, or greater than its initial kinetic energy? Why?

Activity: Energy Transformation on a Curved Ramp and Friction Track

Learning Focus

- Identify which track zones conserve total mechanical energy and which zones drain it.
- Observe how gravitational potential energy converts into kinetic energy during a curved descent.
- Examine how the coefficient of friction and the rough surface length determine whether the block stops on the track or overshoots.

Interactive Applet



Key Observations

- Along the smooth curved ramp, total mechanical energy remains constant while potential energy drops and kinetic energy rises.
- On the horizontal purple track, mechanical energy drops. The loss is exactly equal to the negative work done by kinetic friction.
- Depending on your slider settings, the block will either come to a complete stop on the rough surface or exit with remaining kinetic energy if the track is too short or too smooth.

Challenge / Question

Does changing the curvature or steepness of the frictionless ramp alter the block's velocity at the exact moment it steps onto the flat friction track? Why or why not?

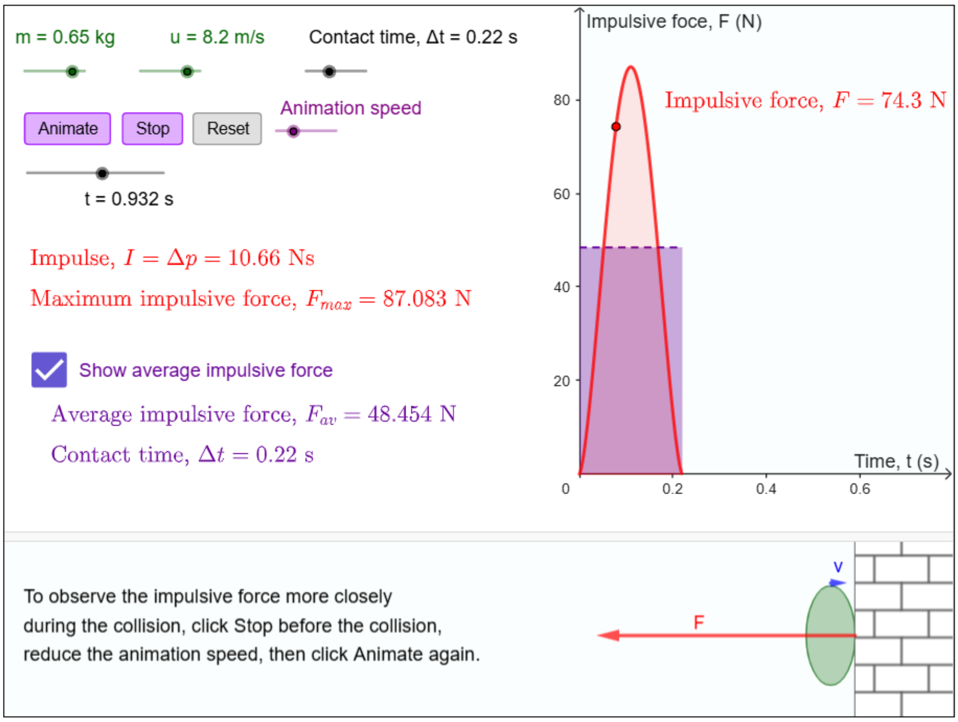
Momentum and Center of Mass

Activity: Impulse, Collision Time, and Impulsive Force

Learning Focus

- Observe how the actual contact force varies dynamically over time rather than remaining constant during a collision.
- Investigate how increasing or decreasing the duration of a collision affects the peak force experienced by an object.
- Observe what physical quantity the total area under the force-time curve represents, and how "average force" simplifies this profile.

Interactive Applet



Key Observations

- The impulsive force is not instantaneous; it builds to a maximum peak as the object deforms against the wall and drops back to zero as it rebounds.
- Increasing the collision contact time flattens the bell curve, drastically lowering the maximum force without changing the total impulse.
- The area of the rectangle formed by the average force is exactly equal to the total area under the peak-shaped impulsive force curve.

Challenge / Question

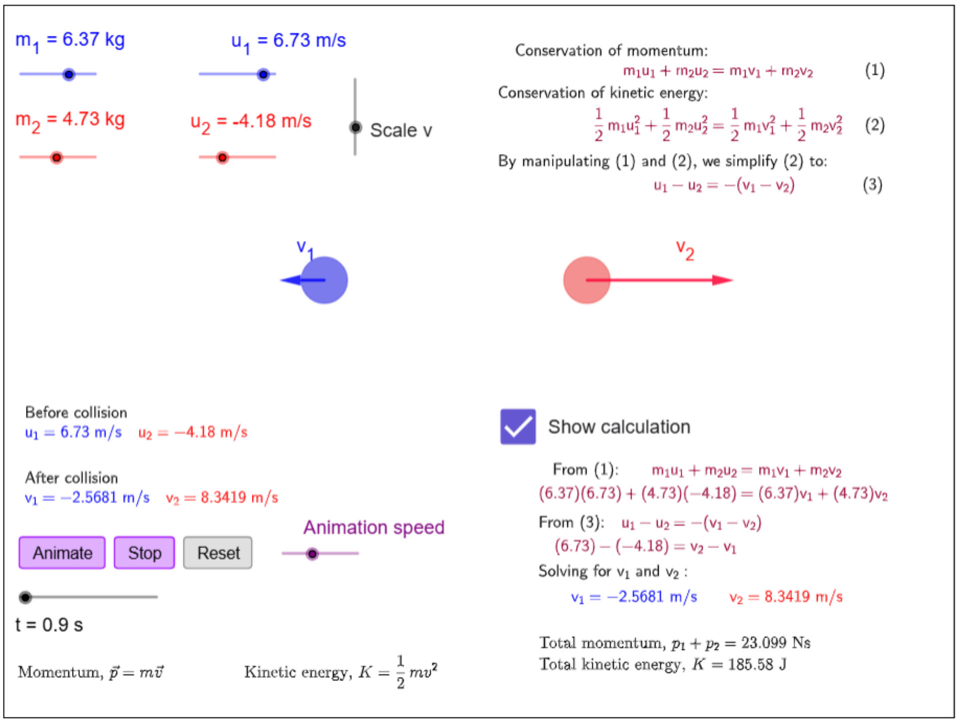
If you adjust the sliders so that a collision has a much longer contact time, how does this explain why cars are designed with "crumple zones" to protect passengers during crashes?

Activity: Conservation of Momentum and Kinetic Energy in One-Dimensional Elastic Collisions

Learning Focus

- Investigate how the mass ratio affects the final speeds and directions of the objects after collision.
- Compare the initial and final velocities of both objects before and after collision.
- Verify that a 1D elastic collision conserves both total momentum and total kinetic energy simultaneously.

Interactive Applet



Key Observations

- If the incoming object is much lighter than the stationary target, it rebounds; if much heavier, it continues forward while pushing the target ahead.
- For equal masses with Object 2 initially at rest, Object 2 leaves with the initial velocity of Object 1, while Object 1 comes to rest.
- Total system momentum and total kinetic energy remain conserved regardless of the mass ratio.

Challenge / Question

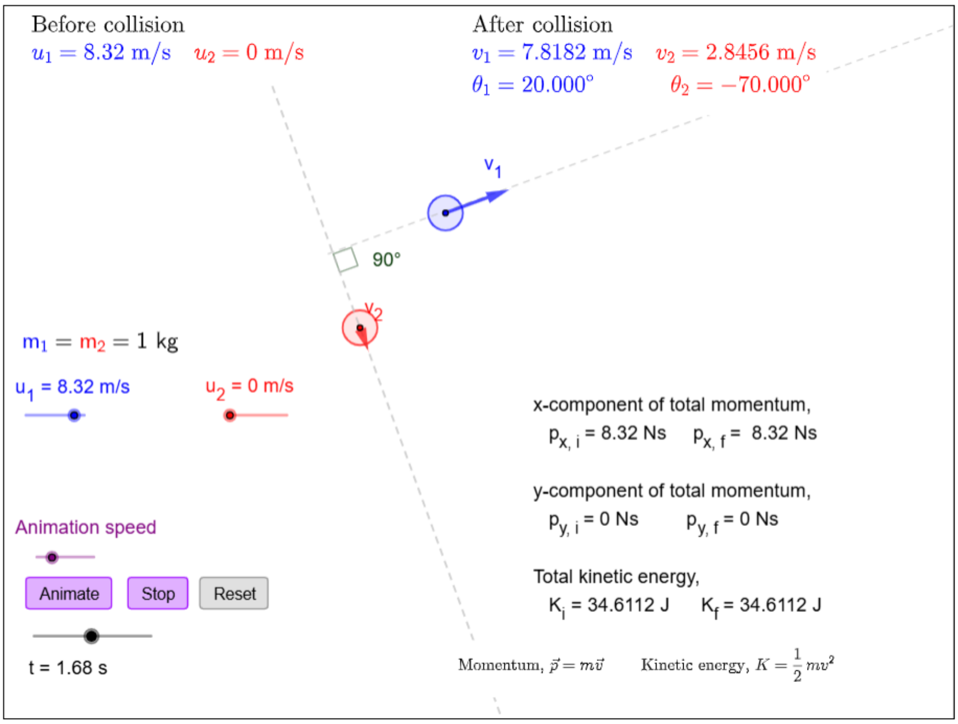
Is it possible to configure both the mass sliders and initial velocity settings so that both objects come to a complete, permanent standstill immediately after hitting each other? Why or why not based on kinetic energy conservation?

Activity: Conservation of Momentum and Kinetic Energy in Two-Dimensional Elastic Collisions

Learning Focus

- Observe how 2D velocity vectors are split into independent horizontal (x) and vertical (y) components to calculate the components of the total momentum.
- Examine how the subsequent scattering angles depend on the glancing nature of the impact, by changing the vertical offset between the centers of the two objects.
- Verify how an elastic collision simultaneously conserves both total momentum and total kinetic energy.

Interactive Applet



Key Observations

- Before and after the collision, the x- and y-components of total momentum of the system remain strictly unchanged.
- Because the interaction is perfectly elastic, the total kinetic energy of the system calculated from the velocity magnitudes remains completely identical before and after impact.
- For a glancing elastic collision between identical masses where one is initially at rest, the objects always scatter at a 90° angle relative to each other.

Challenge / Question

Object 2 is stationary and Object 1 has a zero y-component of velocity (u1y = 0). Since total y-component momentum must remain zero after the collision, how do the final y-components of velocity for Object 1 and Object 2 compare?

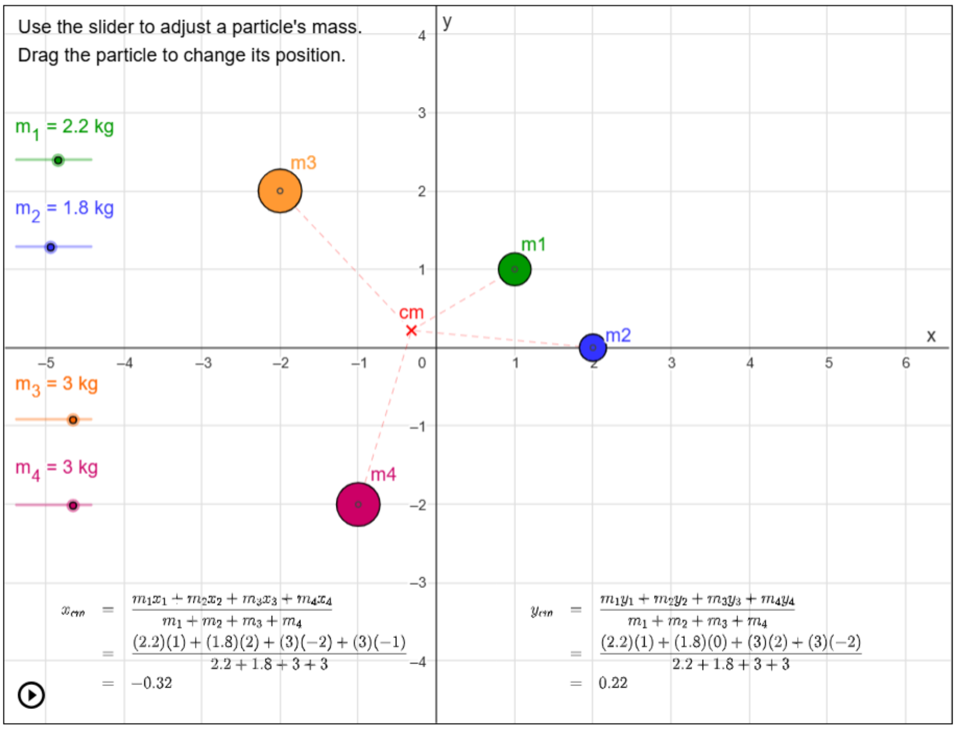
Activity: Calculating the Center of Mass for a System of Particles

Learning Focus

Keep these goals in mind as you explore the simulation:

- Watch how the coordinates and masses from the grid actively populate the numerator and denominator of the x_{cm} and y_{cm} formulas at the bottom.
- Observe how increasing or decreasing a particle's mass slider changes its "pull" on the red cm crosshair, shifting the network of dashed tracking lines.
- Investigate how moving or changing masses can cause one coordinate value (like y_{cm}) to balance perfectly out to zero while the other coordinate remains non-zero.

Interactive Applet



Key Observations

- The center of mass formulas compute a weighted average position. If a system is perfectly balanced symmetrically across an axis, that axis's center of mass calculation evaluates to exactly zero.
- The red cm crosshair is always pulled toward the coordinates containing the highest concentration of mass, shortening the dashed lines leading to heavier particles.
- Regardless of where a particle is moved on the grid, its mass value is always added to the bottom denominator ($m_1 + m_2 + m_3 + m_4$), representing the total combined mass of the entire system.

Challenge / Question

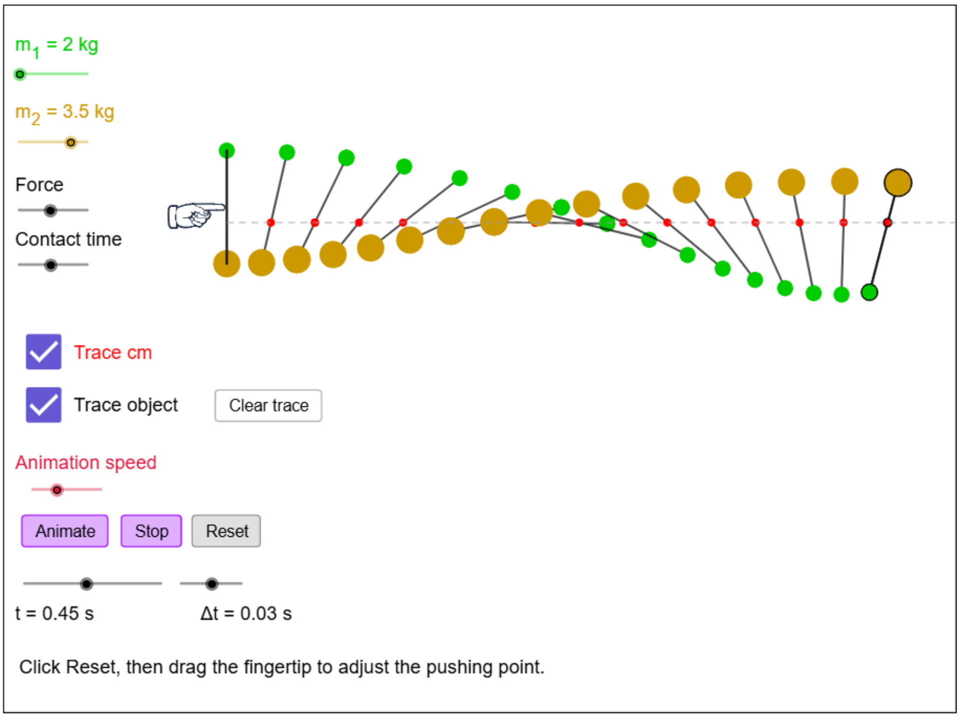
If you place two particles of identical mass symmetrically across the x-axis (same horizontal position, but opposite vertical positions), what happens to the numerator of the y_{cm} calculation box?

Activity: Translational vs. Rotational Motion of an Extended Object

Learning Focus

- Observe how the location where the force is applied along the rod dictates whether the object purely translates or simultaneously rotates.
- Track the trajectory of the red center of mass (cm) dot under different pushing scenarios, regardless of how the rod spins around it.
- Investigate how altering the mass sliders changes the location of the center of mass along the rigid link connecting the two spheres.

Interactive Applet



Key Observations

- If the force is applied exactly at the object's center of mass, the rod moves forward along a perfectly straight line path without any rotation or spinning.
- Applying a force off-center (at any pushing point away from the center of mass) generates a torque, causing the rod to tumble and spin dynamically around its moving center of mass.
- Even when the individual spheres trace complex, wavy, or looping path lines through space, the red center of mass point travels along a clean trajectory.

Challenge / Question

If you make one sphere significantly heavier than the other, the red cm crosshair shifts toward the heavier side. If you want to move the rod forward without making it spin, where must you position the pushing fingertip now?

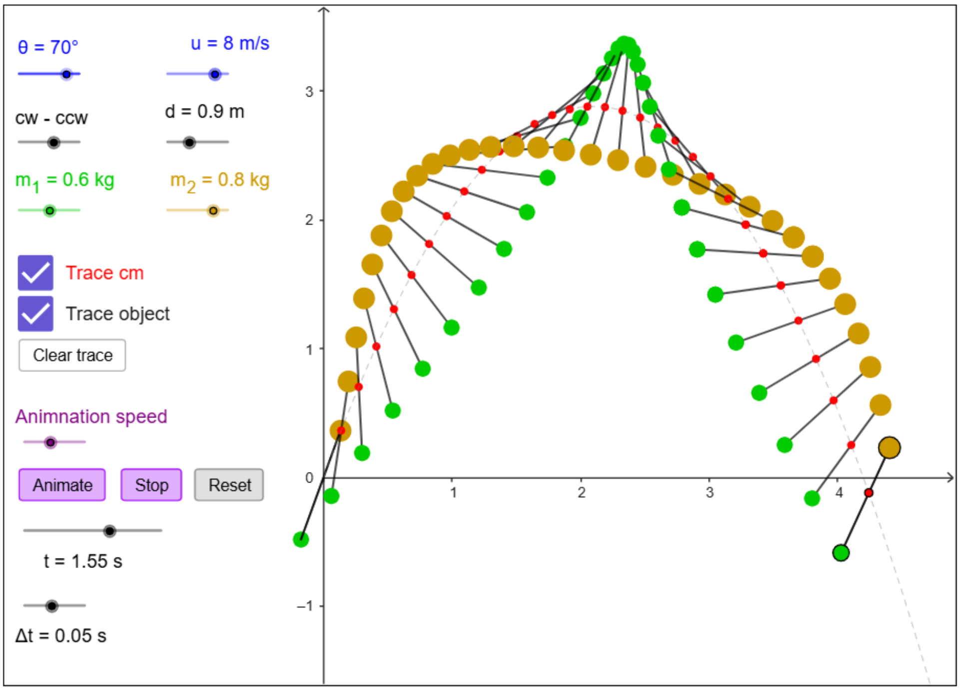
Activity: Projectile Motion of an Extended Object

Learning Focus

Keep these goals in mind as you explore the simulation:

- Observe how a thrown object simultaneously undergoes independent translational projectile motion and rotational motion.
- Focus on the path of the red cm marker compared to the paths of the individual weighted ends of the object.
- Investigate how changing the mass sliders alters the position of the center of mass along the connecting rod and how that affects the flight path visuals.

Interactive Applet



Key Observations

- Even though the individual masses tumble, loop, and track complex wavy paths through the air, the red center of mass point follows a perfectly smooth, unbending parabolic curve.
- Gravity acts on the extended system as if all the combined mass were concentrated entirely at that single red center of mass point.
- The extended object rotates at a steady rate around its own moving center of mass, showing that rotational velocity remains uninfluenced by the linear downward acceleration of gravity.

Challenge / Question

When you turn on both "Trace cm" and "Trace object", you see two very different shapes on screen. Why is the center of mass trace a perfect mathematical parabola, while the individual spheres create looping or scalloped paths?

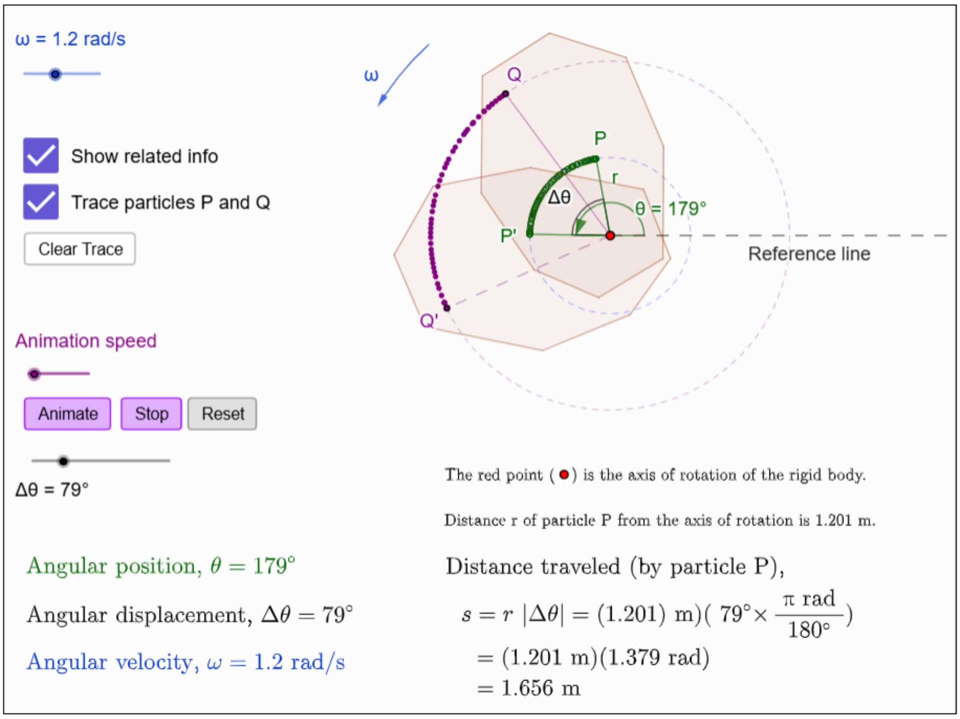
Rotational Motion

Activity: Rotational Kinematics of a Rigid Body

Learning Focus

- Observe how every point within a rotating rigid body shares identical angular quantities while traveling completely different arc paths.
- Track how the numerical values for angular position and angular displacement change for all particles during a rotation at constant angular velocity.
- Investigate how a particle's distance r from the center pivot affects its colored traced path length s.

Interactive Applet



Key Observations

- Every particle on the rigid body sweeps through the exact same angular displacement and rotates with the same angular velocity at any given instant.
- A particle located further from the axis of rotation (larger r) will trace out a longer, sweeping path length (s) than a particle closer to the pivot point.
- Multiplying a particle's radial distance by the shared angular displacement perfectly evaluates to its physically traced path length.

Challenge / Question

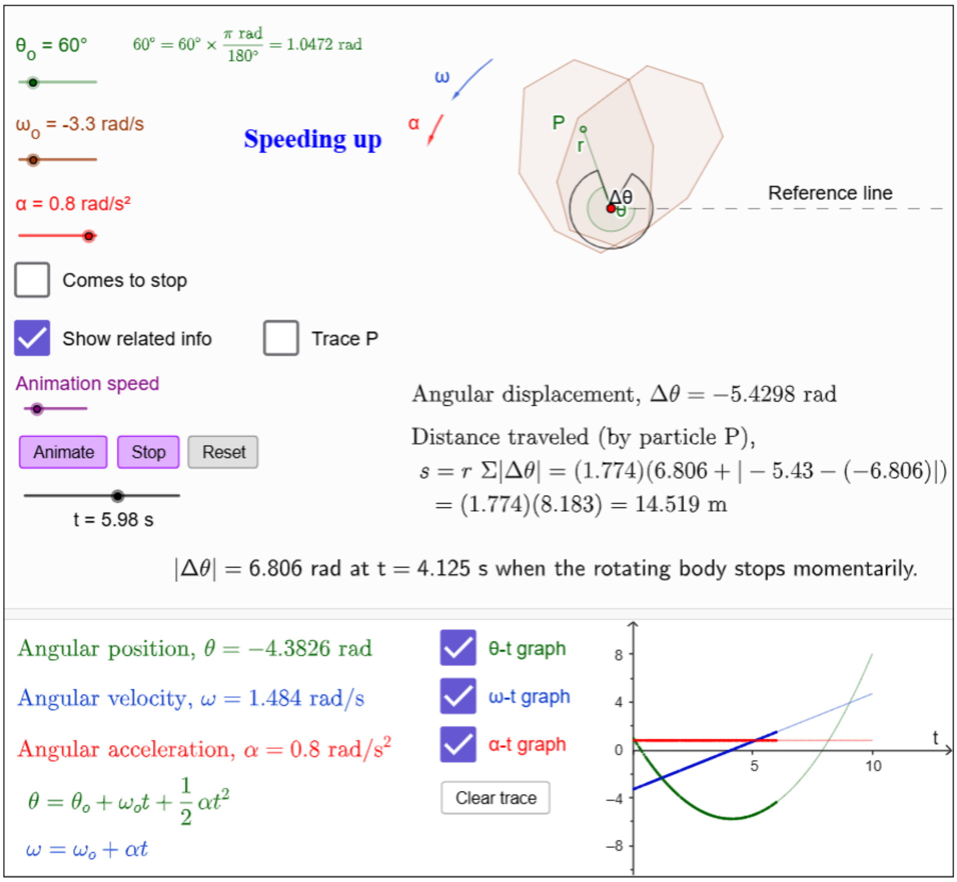
Why do particles P and Q record the exact same angular displacement and angular velocity even though one particle physically traveled a much longer path length across the grid?

Activity: Rotational Kinematics under Constant Angular Acceleration

Learning Focus

- Explore rotational motion under constant angular acceleration.
- Observe how angular position and velocity change over time under a constant angular acceleration and relate them to their corresponding kinematic graphs.
- Investigate how the initial angular position, initial angular velocity, and angular acceleration affect the rotational motion and the distance traveled by a point on the rotating body.

Interactive Applet



Key Observations

- Under constant angular acceleration, the angular velocity changes uniformly, and the angular position changes non-uniformly with time.
- The shapes of the θ-t, ω-t, and α-t graphs correspond to the rotational motion of the body.
- The distance traveled by a point on the rotating body depends on both its distance from the axis of rotation and the angular displacement.

Challenge / Question

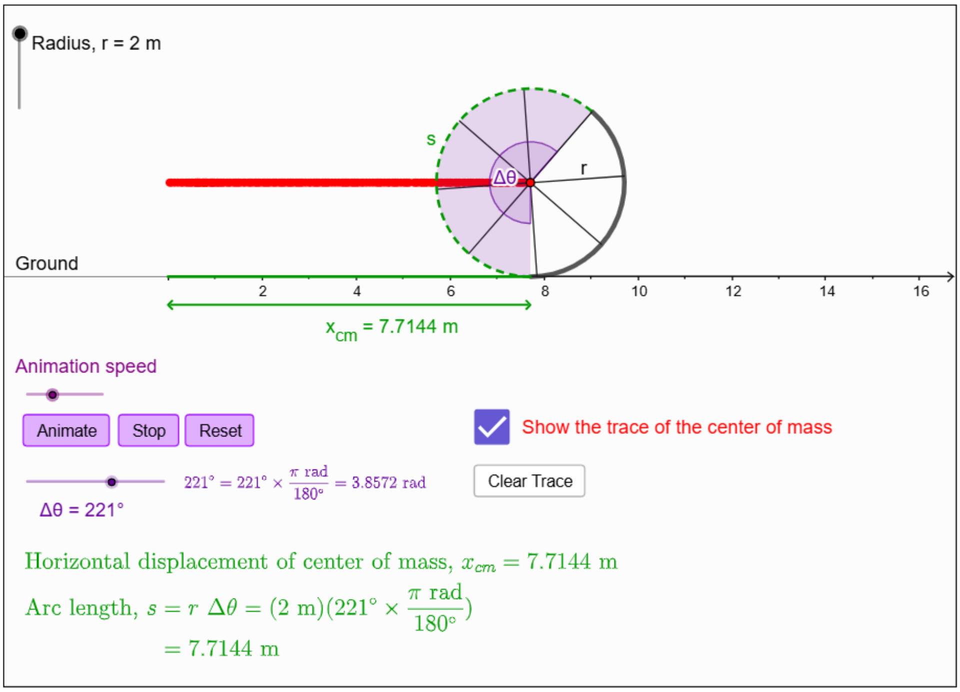
If two points are located at different distances from the axis of rotation, do they have the same angular displacement? Do they travel the same distance? Explain.

Activity: Rolling Motion Without Slipping

Learning Focus

- Observe how the horizontal displacement of the wheel's center of mass tracks identically to the unrolled arc length.
- Notice how degree inputs must scale into radians to calculate linear distances.
- Watch how adjusting the single angular displacement slider updates the linear distance.

Interactive Applet



Key Observations

- x_{cm} = s: In pure rolling, forward translation exactly equals the unrolled outer rim distance.
- Calculating a linear distance ($s = r\ \Delta\theta$) from a rotation requires the angular displacement in radians.
- The wheel's radius (r) acts as the physical bridge converting rotational angles into forward distance.

Challenge / Question

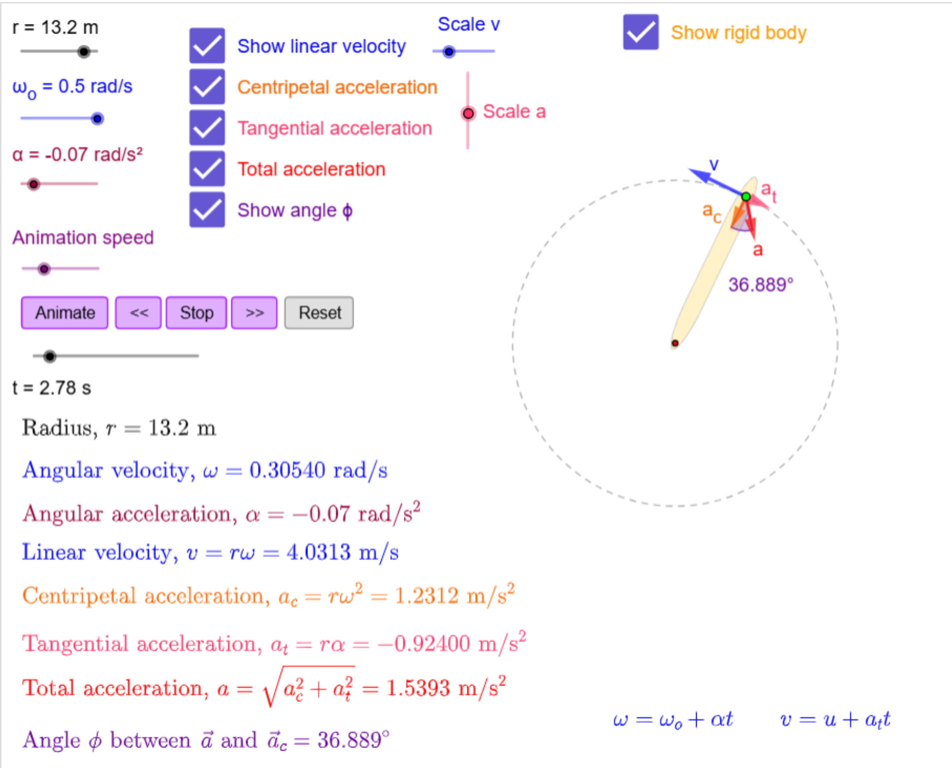
If a tire slips or skids on ice, why would the x_{cm} = s identity break down? Which value would be larger during a burnout?

Activity: Linear and Angular Quantities of a Rotating Rigid Body

Learning Focus

- Explore the relationships between linear and angular quantities in the rotational motion of a rigid body.
- Observe how angular velocity and angular acceleration determine the linear velocity, tangential acceleration, centripetal acceleration, and total acceleration of a point on the rotating body.
- Investigate how the radius, angular velocity, and angular acceleration affect the motion and acceleration of a point on the rigid body.

Interactive Applet



Key Observations

- The linear velocity of a point is always tangent to the circular path, while the centripetal acceleration always points toward the centre of rotation.
- The tangential acceleration depends on both the radius and the angular acceleration, whereas the centripetal acceleration depends on both the radius and the angular velocity.
- The total acceleration is the vector sum of the tangential and centripetal accelerations, and its direction changes as their relative magnitudes change.

Challenge / Question

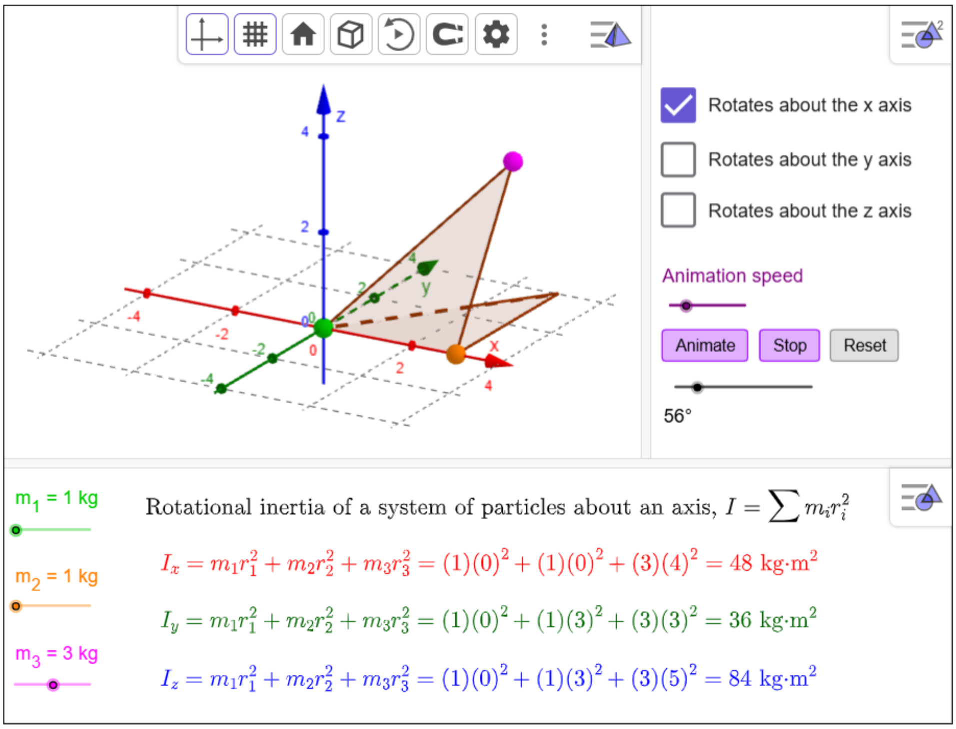
If the angular acceleration is zero but the rigid body continues rotating, which acceleration(s) does a point on the body still experience? Explain why.

Activity: Rotational Inertia of Particles about an Axis

Learning Focus

- Explore the rotational inertia of a system of particles about different axes of rotation.
- Observe how the mass of each particle and its perpendicular distance from the axis of rotation contribute to the rotational inertia.
- Investigate how changing the axis of rotation affects the rotational inertia of the system.

Interactive Applet



Key Observations

- Each particle contributes mr^2 to the rotational inertia, so particles farther from the axis contribute more than those nearer the axis.
- Rotational inertia depends not only on the masses of the particles but also on the chosen axis of rotation.
- Changing the axis of rotation changes the perpendicular distances of the particles from the axis and therefore changes the rotational inertia of the system.

Challenge / Question

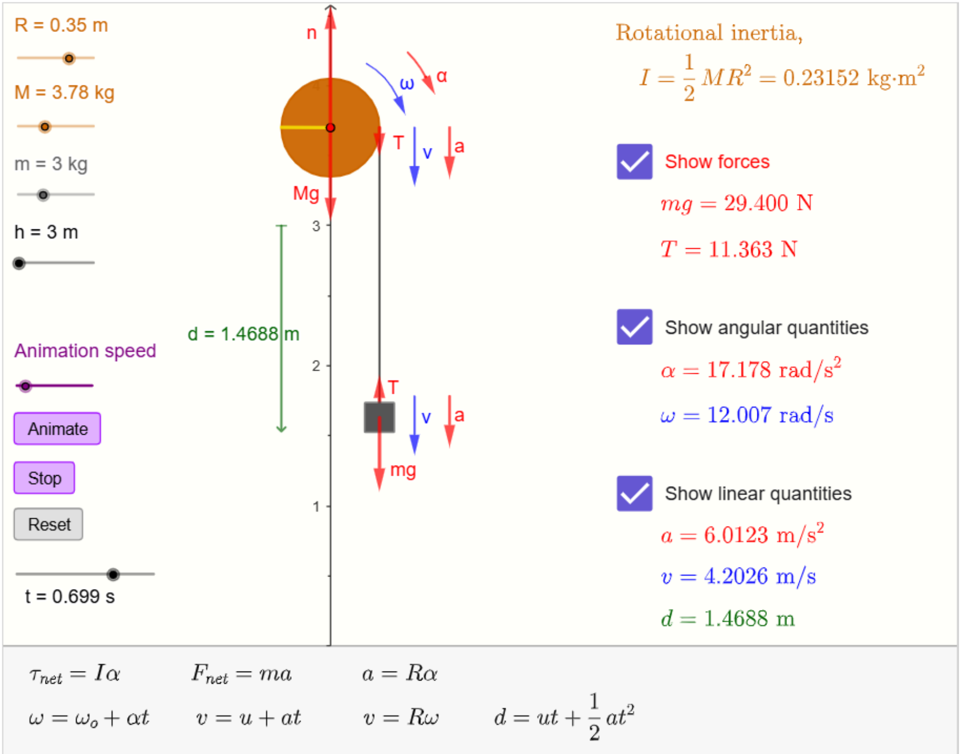
Why can the same system of particles have different rotational inertias about the x-, y-, and z-axes?

Activity: Newton's Second Law in Rotational and Linear Motion

Learning Focus

- Explore the relationship between the linear motion of a falling object and the rotational motion of a pulley.
- Observe how the forces acting on the object and the torque acting on the pulley determine the linear and angular accelerations of the system.
- Investigate how the pulley radius, pulley mass, hanging mass, and initial height affect the motion of the system.

Interactive Applet



Key Observations

- The linear acceleration of the hanging object and the angular acceleration of the pulley are directly related throughout the motion.
- Part of the hanging object's weight accelerates the object, while the remaining part produces the pulley's rotation.
- As the object accelerates downward, its linear velocity and the pulley's angular velocity increase simultaneously.

Challenge / Question

Why is the tension in the string always less than the weight of the hanging object while the object is descending?

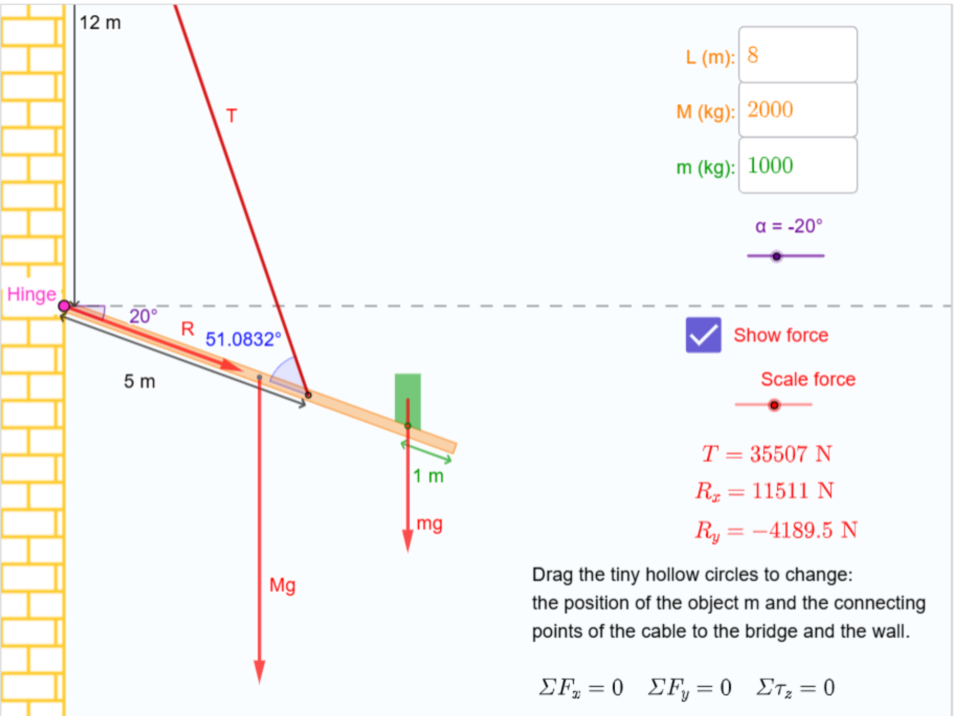
Static Equilibrium

Activity: Bridge with Hinge and Cable Support

Learning Focus

- Explore the conditions required for the static equilibrium of a bridge supported by a hinge and a cable.
- Observe how the cable tension and the hinge reaction force change as the bridge angle, load position, cable attachment position, and masses are varied.
- Investigate how the bridge geometry and the distribution of loads affect the forces and torques acting on the bridge.

Interactive Applet



Key Observations

- The bridge remains in static equilibrium only when both the net force and the net torque acting on it are zero.
- Moving the load farther from the hinge increases the torque due to its weight and generally requires a larger cable tension to maintain equilibrium.
- Changing the bridge angle, load position, or the cable attachment position changes both the cable tension and the hinge reaction force.

Challenge / Question

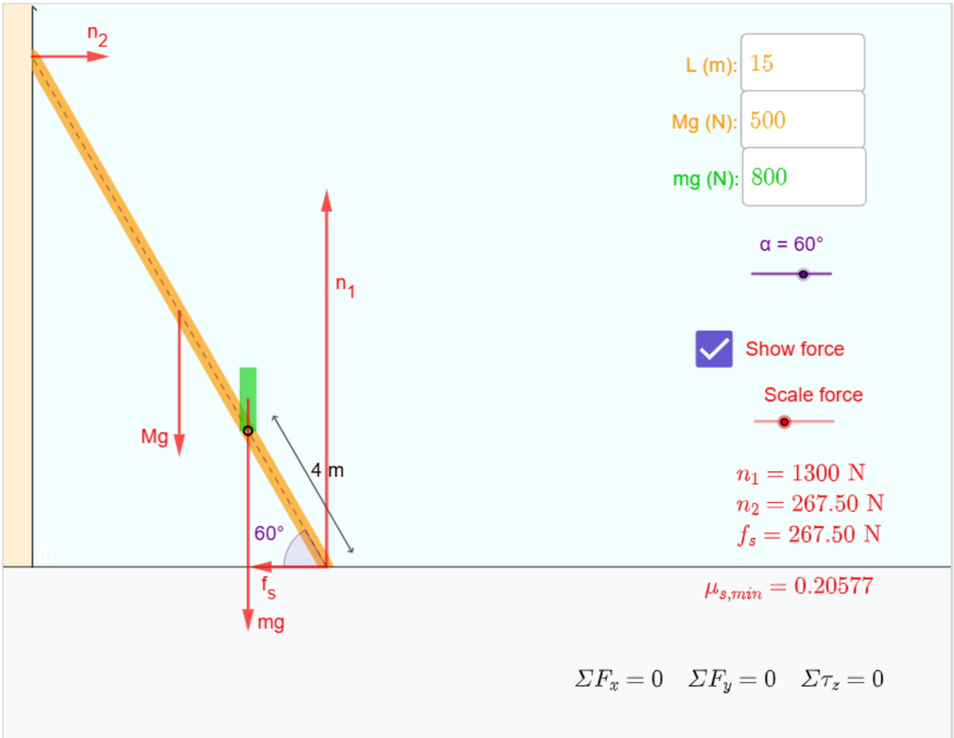
If the cable is attached closer to the hinge, how must the cable tension change to keep the bridge in equilibrium? Explain your reasoning.

Activity: Ladder Leaning Against a Frictionless Wall

Learning Focus

- Explore the conditions required for the static equilibrium of a ladder leaning against a frictionless wall.
- Observe how the ladder angle, ladder weight, person weight, and the person's position affect the normal forces and the static friction required for equilibrium.
- Investigate how the minimum coefficient of static friction changes with the ladder angle and weight distribution.

Interactive Applet



Key Observations

- The ladder remains in static equilibrium only when both the net force and the net torque acting on it are zero.
- As the person moves farther up the ladder, the required static friction at the floor increases.
- Decreasing the ladder angle generally increases the minimum coefficient of static friction required to prevent the ladder from slipping.

Challenge / Question

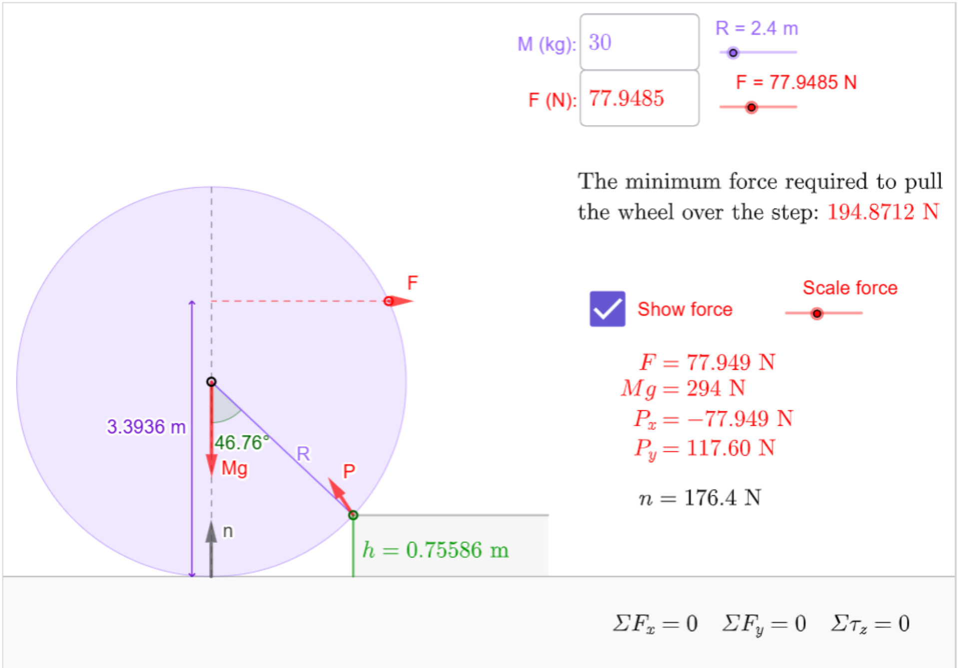
If the floor becomes more slippery (smaller coefficient of static friction), should the ladder be placed at a larger or smaller angle with the floor to remain in equilibrium? Explain your reasoning.

Activity: Minimum Force Required to Pull a Wheel Over a Step

Learning Focus

- Explore the conditions required for a wheel to just begin rotating over a step.
- Observe how the applied force, wheel weight, and contact force at the step determine the critical condition for rotation.
- Investigate how the wheel radius, wheel weight, and step height affect the minimum force required to pull the wheel over the step.

Interactive Applet



Key Observations

- The minimum pulling force occurs when the wheel is about to rotate about the edge of the step.
- Increasing the wheel radius generally reduces the minimum force required, whereas increasing the wheel weight or step height generally increases it.
- At the critical condition, the ground no longer exerts a normal force on the wheel, and the edge of the step becomes the pivot.

Challenge / Question

Why does the edge of the step, rather than the centre of the wheel or the ground, become the pivot when the wheel is just about to climb over the step?

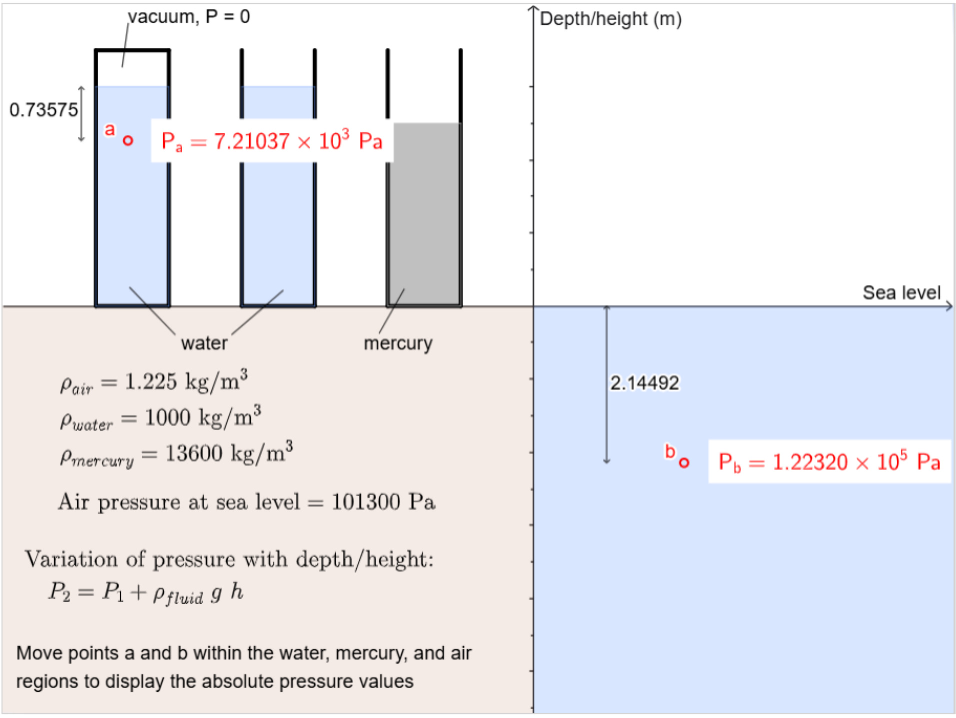
Fluid Mechanics

Activity: Variation of Pressure with Depth and Height

Learning Focus

- Note the reference air pressure at sea level (0 m) is set to standard atmospheric pressure (101,300 Pa).
- Compare how the absolute pressure changes when moving above sea level (into the air) versus moving below sea level (into water).
- Notice how absolute pressure at the same depth increases when switching from water to the denser mercury.

Interactive Applet



Key Observations

- Fluid pressure increases linearly the deeper you submerge below the reference level due to the weight of the fluid above.
- Denser fluids (like mercury compared to water) generate a steeper pressure increase per meter of depth.
- The applet dynamically evaluates absolute pressure using the fundamental fluid relationship: $P_2 = P_1 + \rho_{fluid}\, g\, h$

Challenge / Question

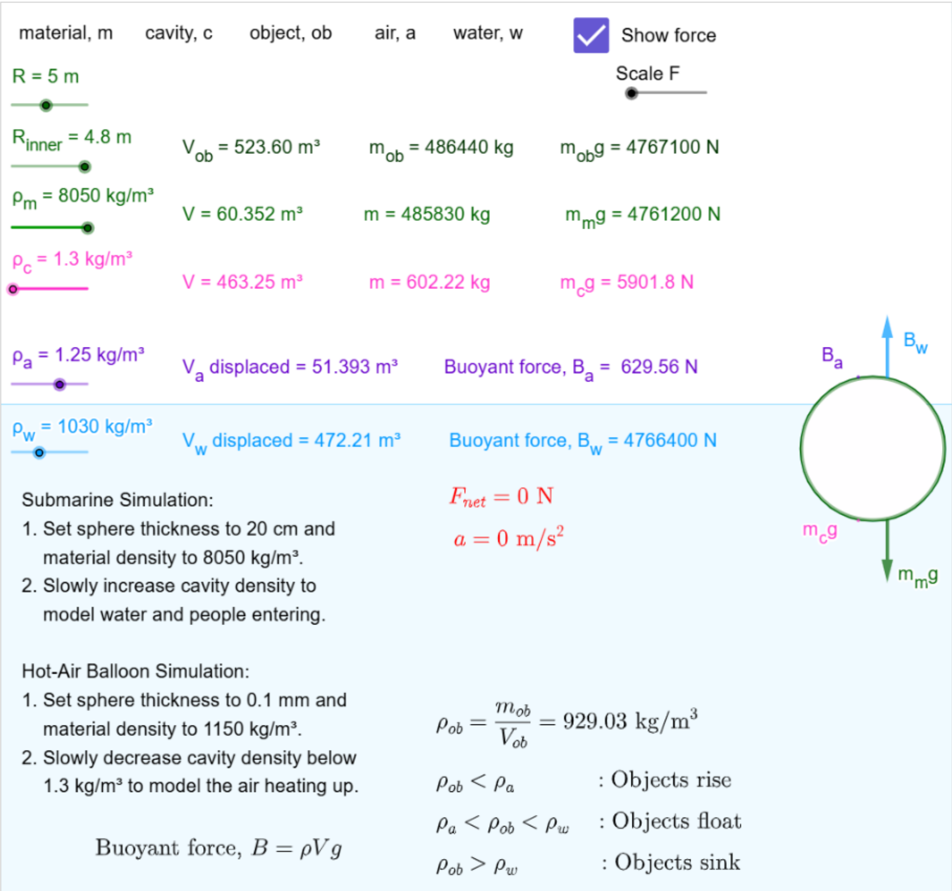
Look at the data panel listing the densities of air, water, and mercury. Why does moving a point 1 meter downward in mercury increase the pressure more drastically than moving it 1 meter downward in water?

Activity: Buoyant Force – Why Objects Sink, Float, or Rise

Learning Focus

- Watch how the average object density (ρ_{ob}) relative to air (ρ_a) or water (ρ_w) dictates whether an object rises, floats, or sinks.
- Notice that the object simultaneously experiences a buoyant force from the air (B_a) and a buoyant force from the water (B_w) depending on its submerged volume.
- Observe how varying the material density ρ_m) and cavity density (ρ_c) changes the object's total mass, thereby altering its overall weight.

Interactive Applet



Key Observations

- Buoyant force exactly equals the weight of the fluid displaced by the object ($B = \rho V g$).
- An object's net behavior is determined by the fluid density thresholds:
  - $\rho_{ob} < \rho_a$: Object rises.
  - $\rho_a < \rho_{ob} < \rho_w$: Object floats in water.
  - $\rho_{ob} > \rho_w$: Object sinks.
- Static floating equilibrium occurs when total upward buoyancy matches total downward weight, resulting in zero net force and zero acceleration.

Challenge / Question

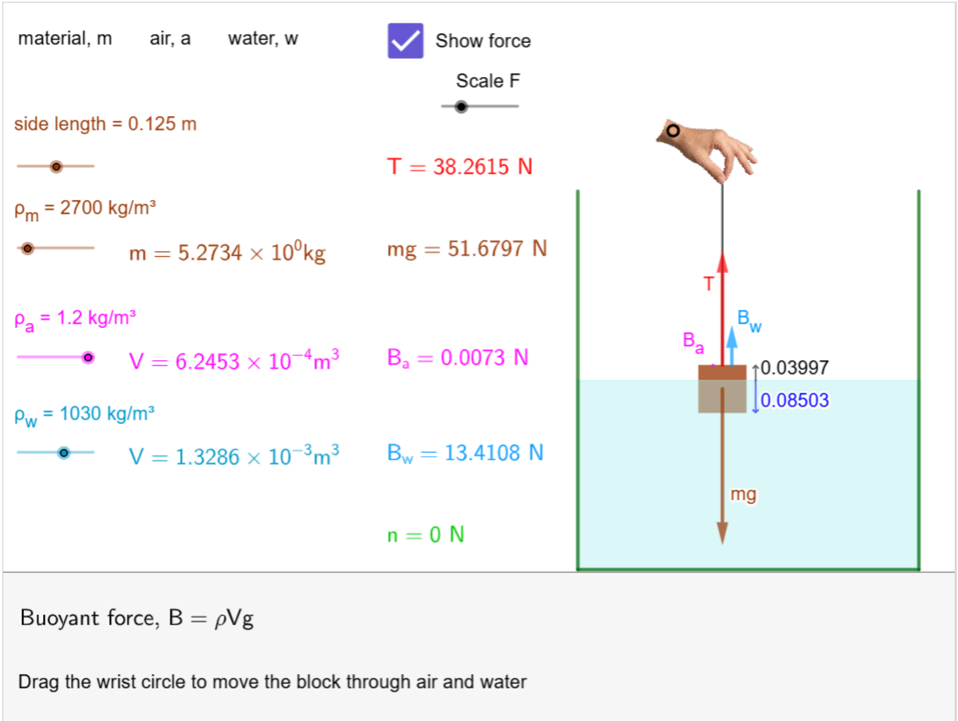
Based on the Submarine Simulation, compare the values of the buoyant force of air (B_a) and water (B_w). Why does this comparison justify ignoring air buoyancy in standard textbook problems?

Activity: Buoyancy and Force Equilibrium

Learning Focus

- Explore how fluid density and submerged volume dictate the magnitude of the buoyant force.
- Investigate the transition of buoyant forces as an object moves from a low-density fluid (air) to a high-density fluid (water).
- Analyze how string tension (T) dynamically adjust to maintain static equilibrium as buoyancy changes.

Interactive Applet



Key Observations

- Air buoyancy is negligible due to its low density, leaving tension to support nearly the full weight. Water buoyancy is vastly larger because water is much denser than air.
- As the object lowers into water, the buoyant force increases linearly with the submerged volume, causing a matching linear drop in string tension.
- Once fully submerged, buoyancy plateaus. If the object's density exceeds the fluid's density, the remaining weight must be supported either by string tension (when suspended) or by the normal force (when resting on the bottom).

Challenge / Question

What exact relationship must exist between the material density and the fluid density for the block to hover perfectly suspended in the water with zero string tension? Explain your reasoning.

Acknowledgements

The **GeoGebra Interactive Physics Fundamentals Framework (GIPFF)** was developed using **GeoGebra**, whose powerful mathematical and visualization tools make interactive physics learning possible.

The concepts presented in this framework are primarily based on the following reference books:

- Physics for Scientists and Engineers — Raymond A. Serway, John W. Jewett
- Physics for Scientists and Engineers — Douglas C. Giancoli
- University Physics — Hugh D. Young, Roger A. Freedman, A. Lewis Ford
- Fundamentals of Physics — David Halliday, Robert Resnick, Jearl Walker

The author gratefully acknowledges the contributions of the GeoGebra community and the authors of these textbooks, whose work has inspired the development of this framework.

GIPFF is a free educational resource intended to support the teaching and learning of fundamental physics.